

Car aerodynamics.

From first principles to a better package.

An engineer's four-day course in understanding airflow, creating downforce, controlling drag, and making decisions from evidence.

4 days · 28 guided hours · 24 lessons · 12 practical labs · Start with no aero background

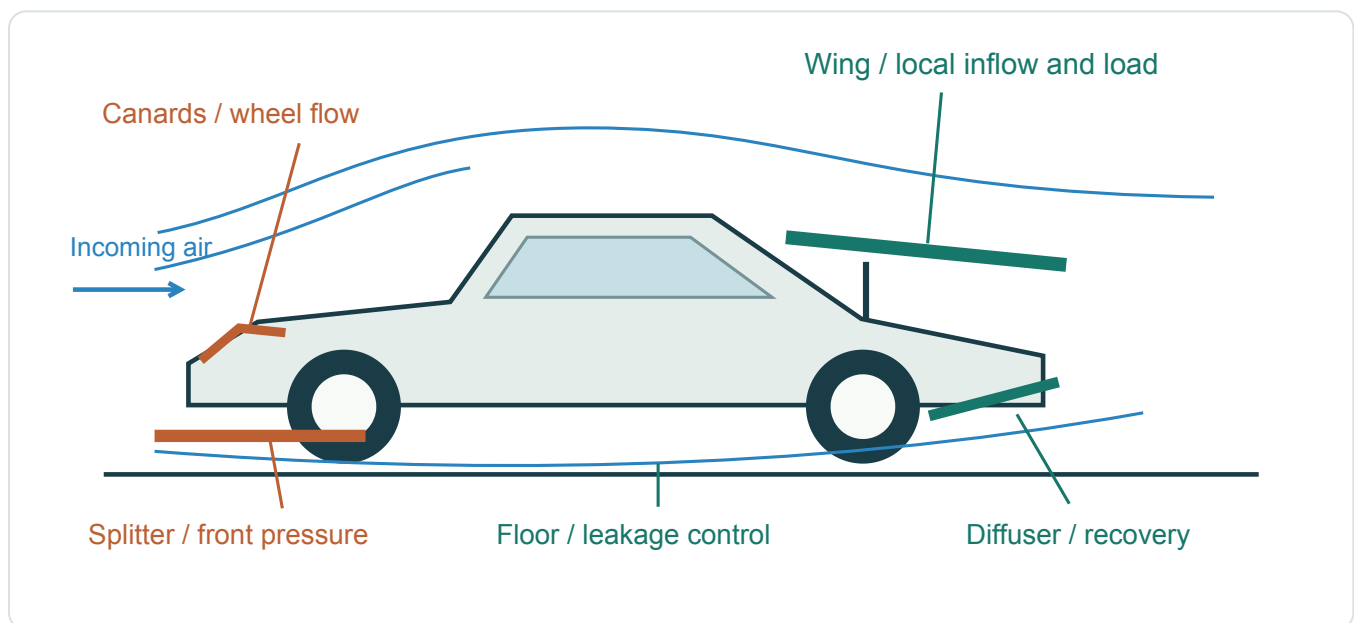


Figure 1. The aerodynamic package is a connected flow system. Arrows are a schematic, not a computed flow field.

READING MAP

Course contents

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23. Day 4 Physical validation and a buildable modification
24. Day 4 Capstone: a defensible aerodynamic package

Appendices: worked answers; glossary and formula sheet; lab record; sources. PDF bookmarks also follow the lesson headings.

BEFORE YOU BEGIN

A course you can work through, not just read

You already know how to reason about force, energy, and uncertainty. This course builds the missing aerodynamic intuition: how a car changes pressure and momentum, why flow separates, how devices interact, and how to decide whether an apparent improvement is trustworthy. By the end you should be able to write a testable design hypothesis, prepare comparable simulations, explain their limitations, and specify the physical evidence needed before building a part.

Allocate seven hours per day, excluding lunch. Each day contains six lessons and three labs. A suggested day is 09:00–10:30 reading and derivation, 10:45–12:15 devices or methods, 13:00–15:00 labs, and 15:15–17:15 review, calculations, and discussion. The twelve labs, including the capstone, account for roughly fourteen hours; allow ten for reading and derivation and four for checks, debriefs, and interpreting figures. On Day 4 combine the two afternoon blocks for the four-hour capstone. A solver taking longer does not shorten the learning objective: use the included run evidence while a repeat computes.

DAY 1

Read the flow

Forces, pressure, scaling, boundary layers, wakes, and measurement.

DAY 2

Understand the devices

Wings, splitters, canards, floors, diffusers, and their operating envelope.

DAY 3

Build useful experiments

A research case study, CFD setup, convergence, comparisons, and diagnosis.

DAY 4

Design the whole car

Balance, speed tradeoffs, interactions, robustness, testing, and a capstone.

Three kinds of evidence appear in this course

Schematic

Original drawings explain a mechanism or coordinate convention. A streamline drawn by hand is not a solution of the governing equations.

Teaching calculation

Calculators evaluate stated equations under stated assumptions. Their output is not a prediction for your car.

CFD / literature

Actual EasyCFD run output and the supplied research paper retain their own geometry, settings, and qualification limits. Read the caption before interpreting a colour.

Your tools and your notebook

The reading, diagrams, calculators, answers, and progress work inside this HTML file without a backend or WebGPU. Download a copy using the toolbar. The app exercises need EasyCFD and a WebGPU-capable browser; local OpenFOAM exercises additionally need the existing Docker/backend setup. The public edition

offers WebGPU; it does not run OpenFOAM on demand. A calculator continues to work even when a CFD solver is unavailable. Reading progress and notes are saved locally when browser storage is available; export notes before moving to another device.

For each lab record four things: your prediction before running, the controlled settings, the observation, and the interpretation. Write down what would disprove your explanation. Keep forces in newtons, geometry in metres, and speed in metres per second in your calculations. Use the fixed car reference area of 2.2 m^2 for the teaching pack. That is an intentionally declared reporting area, not a claim about the frontal area of another vehicle.

Lab 01 · 30 min · Establish your engineering question

Choose one car and one use: road efficiency, high-speed stability, a fast circuit, or a slow circuit. State the speeds, mass, tyre constraints, cooling needs, allowable dimensions, and the behaviour you want to change. Write an acceptance statement such as “increase front aerodynamic load at 140 km/h while keeping added aerodynamic power below 4 kW at 200 km/h.” The numerical limits are your design requirements, not universal recommendations.

Deliverable: a one-page brief, a baseline photograph or geometry reference, and a list of five unknowns. Do not pick a wing angle yet. A device is a proposed answer to the brief, not the starting question.

DAY 1 / 01 / FOUNDATIONS

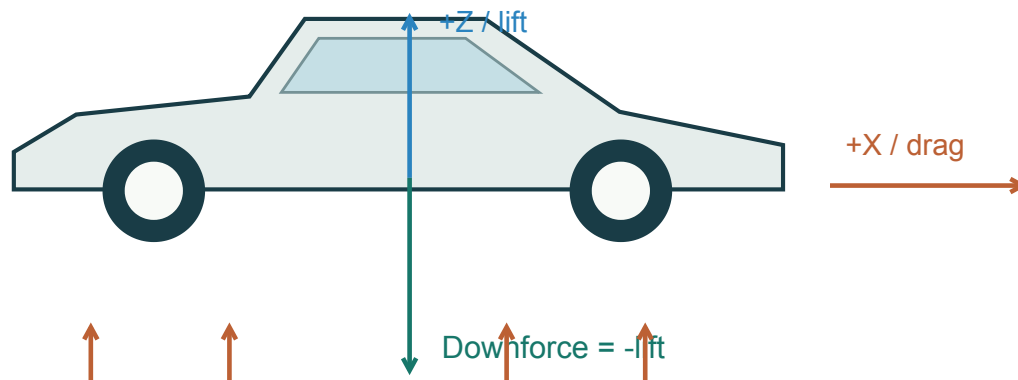
Forces, axes, and the design objective

You will distinguish drag, lift, downforce, and moments, and translate a design goal into measurable quantities.

Air loads every exposed surface

Air exerts pressure normal to a surface and viscous shear along it. Add those small forces over the complete exposed surface and you obtain the total aerodynamic force. Air also produces a moment because those forces act at different positions. A wing is therefore not the only source of downforce: the body, floor, wheels, and cooling flow all contribute. Conversely, adding a part changes the pressure on existing surfaces, so the total car change can differ substantially from the force on the new part.

In a car-fixed view the air approaches the nose and moves toward the rear. EasyCFD uses a nose pointing toward $-X$, airflow toward $+X$ at zero yaw, and $+Z$ upward. Positive lift is upward force. Define positive downforce as $N_{\text{aero}} = -L$. If the body and wheels make $+400$ N lift and a wing contributes -150 N in the same installed flow, the whole-car total is $+250$ N lift. That remaining lift still unloads the tyres. A negative downforce number is uplift under this convention.



Pressure acts normal to each surface; shear acts tangentially.

Figure 2. Surface pressure, shear, and the integrated force. The coordinate arrows follow EasyCFD. Force arrows are illustrative.

$$\mathbf{F} = \int_{\text{surface}} [-p \mathbf{n} + \boldsymbol{\tau} \cdot \mathbf{n}] dS$$

$$\mathbf{M}_O = \int_{\text{surface}} (\mathbf{r} - \mathbf{r}_O) \times d\mathbf{F}$$

Here $\mathbf{n}$ is the outward normal from the solid into the air, p is pressure, $\boldsymbol{\tau}$ is the viscous stress tensor, and $\mathbf{O}$ is the chosen moment origin. Uniform ambient pressure cancels on a closed surface. That cancellation is one reason holes, missing faces, and incorrectly counted internal surfaces can corrupt an integration. Gauge pressure is convenient, but the same reference must apply everywhere you integrate. Pressure on the underside also pushes into the solid; whether it contributes upward or downward depends on both pressure relative to ambient and the surface normal.

Which improvement do you actually want?

A road car may benefit from reducing drag while keeping axle lift within acceptable bounds. A track car may accept more drag for useful downforce, provided the balance, ride-height sensitivity, and structural loads are acceptable. There is no single “more aerodynamic” score. Minimising drag with no constraint can lead to a body that lifts. Maximising downforce with no constraint can produce an unusable straight-line power demand. The engineer's task is to find a feasible package for the operating conditions.

Keep three metrics separate: total drag D , total positive downforce N_{aero} , and the front share of that downforce. Also track side force and yawing moment in crosswind. The ratio N_{aero}/D is an efficiency measure, but a very efficient device that adds almost no load may not meet your target. The incremental ratio $\Delta N_{\text{aero}}/\Delta D$ describes a modification only when ΔD is positive and meaningfully different from zero. Near zero added drag, report the two changes separately; an enormous ratio can hide numerical noise.

Pressure is not weight

A force of 981 N is approximately 100 kgf, meaning the force equivalent of the weight of 100 kg under standard gravity. It does not add 100 kg of vehicle mass or inertia. Aero load disappears at low speed, changes in wind, and may move between axles. Use newtons for mechanics and label any kilogram display as an equivalent load at a specified speed. You cannot use that number alone to specify suspension, tyre grip, or structural strength.

Worked example

A baseline has $D = 420$ N and $L = +180$ N. A package has $D = 500$ N and $L = -120$ N at the same speed. Added downforce is $-(-120) - (-180) = 300$ N, added drag is 80 N, and incremental efficiency is 3.75. The package's total downforce is only 120 N. Mixing those two downforce quantities exaggerates the final load.

▼ Check your understanding: a wing makes 200 N downforce, but the car has 100 N lift. Is the car pushed onto the road?

No. The wing's negative lift is included in the total. Total $L = +100$ N still unloads the tyres. You need the whole-car force and moment, not just the wing result.

Reference for coefficient and axis declarations: OpenFOAM forceCoeffs documentation. The equations and examples here are original teaching material.

DAY 1 / 02 / FOUNDATIONS

Pressure, momentum, and Bernoulli

You will explain a pressure field without relying on the equal-transit-time myth or treating Bernoulli as a universal solver.

Follow a parcel, then follow the forces

Air changes velocity because forces act on it. A pressure gradient is one of those forces; viscosity is another. A curved streamline requires acceleration toward its centre of curvature and therefore a corresponding force. Around a car, geometry and the boundary conditions establish a connected velocity and pressure field. You cannot decide that the air must move twice as fast under a car just because a local gap halves. The amount of air entering the gap can also change, and flow can escape sideways.

Conservation of mass in a steady streamtube gives $\rho U A = \text{constant}$ when velocity is represented by an appropriate area average. For constant density, a smaller cross-section requires higher mean speed *if the same mass flow stays in that streamtube*. A car's floor is an open three-dimensional system, unlike a sealed pipe. The streamtube boundaries themselves depend on pressure, leakage, wheels, and yaw. The continuity equation is a constraint on a solution, not permission to prescribe a solution from a sketch.

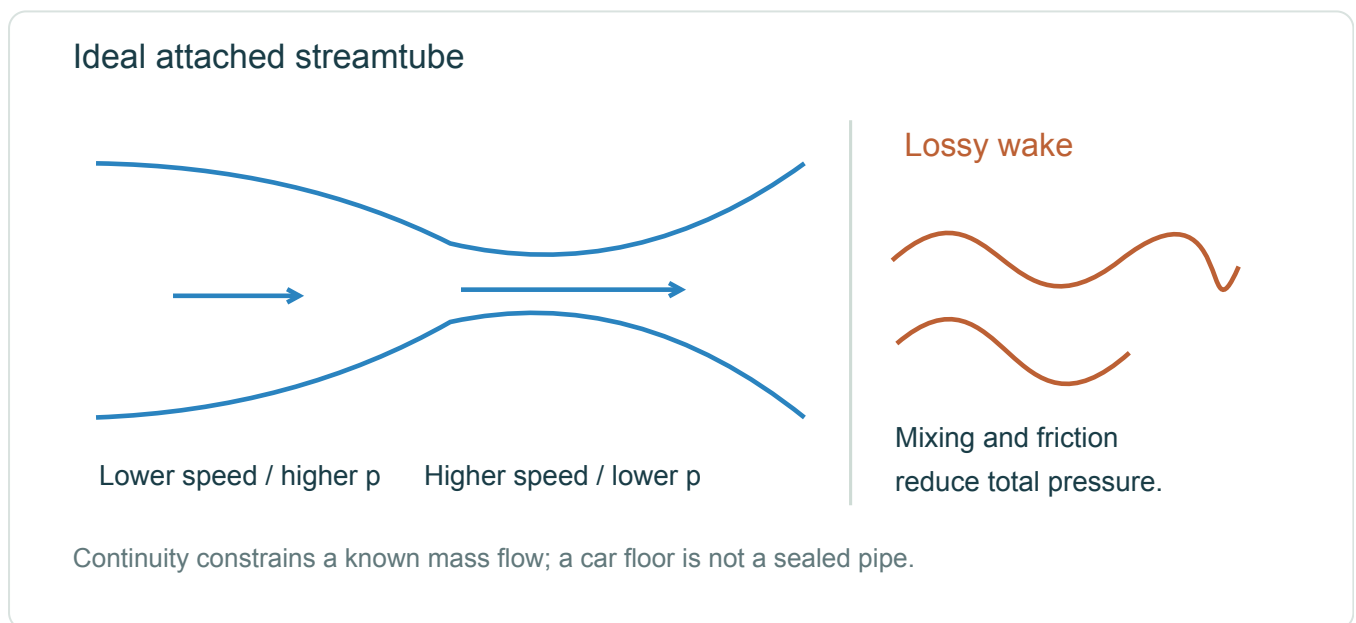


Figure 3. An ideal streamtube exchanges static and dynamic pressure. A real separated wake loses total pressure. This is a schematic of the distinction.

$$p + \frac{1}{2}\rho U^2 + \rho g z = \text{constant along a streamline}$$

$$C_p = (p - p_\infty) / q_\infty \quad ; \quad q_\infty = \frac{1}{2}\rho U_\infty^2$$

The first expression is the steady, inviscid, incompressible Bernoulli relation without work or heat input. At car scales the elevation term in air is usually small compared with dynamic pressure. Outside strongly viscous regions, along a streamline from a common freestream, it provides a useful way to relate local pressure and speed. It cannot restore energy lost to wall friction and mixing, predict where a diffuser will separate, or determine mass flow through a radiator. Streamlines coming from different sources need not have the same total pressure.

Stagnation and suction

At a well-behaved incompressible stagnation point, local speed approaches zero and static pressure approaches freestream total pressure; C_p is approximately +1. A value of $C_p = -0.6$ denotes pressure 0.6 dynamic-pressure units below freestream static pressure. It is not a negative absolute pressure. With $q = 500$ Pa the gauge pressure is -300 Pa, still very close to the roughly 100,000 Pa ambient absolute pressure. Suction on the underside can pull a floor downward; suction on a roof commonly pulls the roof upward. The colour alone does not determine the force direction.

A turning flow and a pressure integral are consistent descriptions of the same interaction. A downward-loaded wing pushes air upward on average; the reaction on the wing is downward. The detailed pressure distribution is how that reaction is transmitted to the solid. Calling one explanation “Newton” and the other “Bernoulli” does not make them competing mechanisms. A correct flow satisfies momentum conservation as well as its applicable energy relation.

What a wing does not require

Air parcels that separate at a leading edge are not obliged to meet again at the trailing edge at the same time. An inclined flat plate can generate lift without two different path lengths. A symmetric airfoil can generate lift at incidence. In a useful attached-flow model, circulation, trailing-edge behaviour, camber, and angle of attack jointly describe the velocity field. Viscosity matters even when most of the integrated lift comes from pressure because it governs boundary layers, separation, and the selected flow state.

Worked example

At $U_\infty = 30$ m/s and $\rho = 1.225$ kg/m³, $q_\infty = 551.25$ Pa. Ideal acceleration to 36 m/s gives $C_p = 1 - (36/30)^2 = -0.44$ and a static-pressure change of -242.55 Pa. This calculation assumes unchanged total pressure along that streamline. In a lossy floor passage, the same C_p cannot be used to infer speed without accounting for the loss.

▼ Check: can a larger blue low-pressure patch on the roof mean more downforce?

Not by itself. Lower pressure on an upward-facing roof often increases upward force. Integrate pressure with the surface normals over the complete geometry and include shear.

Further reading: NASA's Bernoulli equation and its restrictions. Use the equation as a limited ideal model; the car examples are derived here.

DAY 1 / 03 / FOUNDATIONS

Speed, coefficients, and similarity

You will convert coefficients into forces and power, and decide which dimensionless groups matter.

Dynamic pressure is the natural force scale

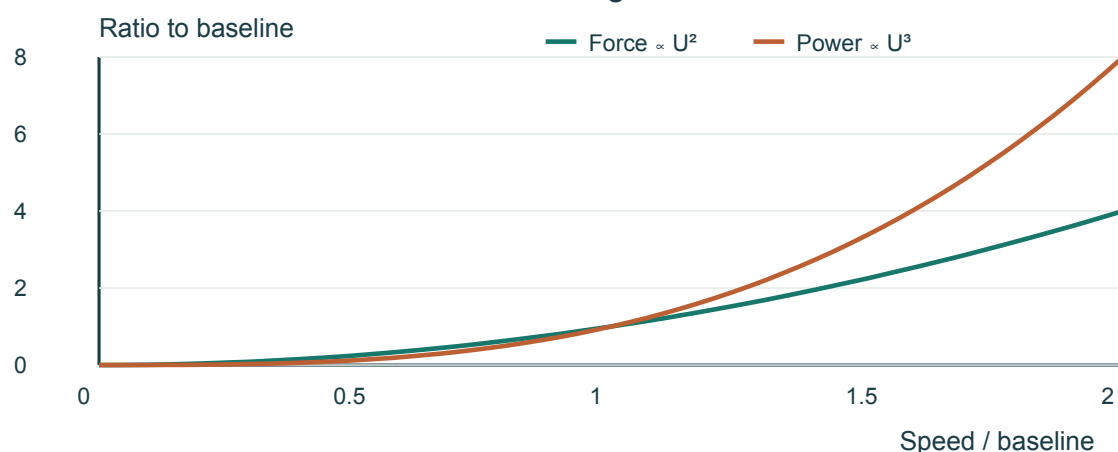
Define $C_D = D/(q\infty A_{ref})$ and $C_L = L/(q\infty A_{ref})$. A coefficient removes the explicit density, speed-squared, and area scale, making comparison easier. It does not remove every dependence on speed: Reynolds number, ground clearance, yaw, and flow state can still change the coefficient. A declared reference area is part of the result. Two C_d values using different areas cannot be compared directly. The product $C_d A$, often called drag area, is particularly useful for comparing cars of different size.

The usual whole-car reporting area is frontal area; a wing commonly uses planform area. Converting is simple: $C_{d,car} = C_{d,wing} \times S_{wing}/A_{car}$ for the same dimensional wing drag and freestream q . If the local wing sees a different dynamic pressure, use its actual force first and then divide by the car's reference q and area. Never insert a car-area lift coefficient directly into a wing's induced-drag formula.

$$D = \frac{1}{2}\rho U^2 C_D A \quad ; \quad L = \frac{1}{2}\rho U^2 C_L A$$

$$P_{aero} = D U \text{ in still air}$$

Constant coefficients / normalised scaling



At twice the speed: four times the force, eight times still-air aero power.

Figure 4. Constant-coefficient forces grow with U^2 ; still-air aerodynamic power grows with U^3 . Normalised curves are teaching calculations.

Workbench · Force and power

Teaching calculation: constant coefficients, still air, no change in ride height or separation. This does not run CFD.

Road speed (still air)	100 km/h	Drag coefficient C_d	0.32
Signed lift coefficient C_l	-0.4	Car reference area	2.2 m²
Air density	1.225 kg/m³		

472.6 Pa

Dynamic pressure

332.7 N

Drag

415.9 N

Downforce (negative = uplift)

9.24 kW

Still-air aero power

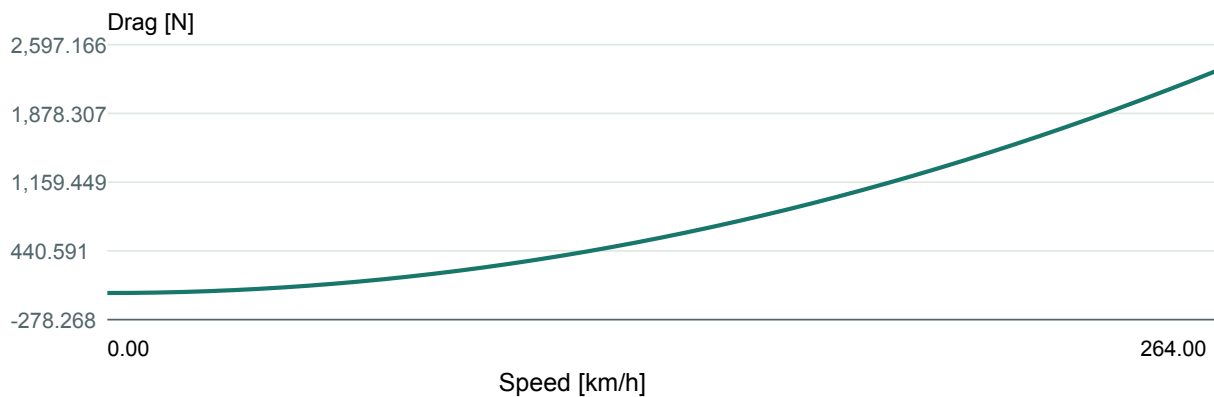
7.78 million

Car-length Reynolds number

0.081

Mach ($a = 343$ m/s)

Constant- C_d drag curve



Reynolds number: inertia versus viscosity

$Re = UL/v$, where v is kinematic viscosity and L is a declared length. Use car length for a global comparison and wing chord for section behaviour. At 100 km/h, $L = 4.2$ m and $v = 1.5 \times 10^{-5}$ m²/s give $Re \approx 7.78$ million. The same air over a 0.32 m wing chord has $Re \approx 593,000$. The wing's surface roughness, transition, and laminar separation may therefore matter differently from those of the body. A reduced-scale wind-tunnel model does not automatically reproduce the full-size car.

A quarter-scale model tested at the same speed has one quarter of the length-based Reynolds number. Matching it in the same air would require four times the speed, which may introduce significant compressibility, change facility limits, and increase loads. Transition trips can deliberately set boundary-layer state in a model test, but they are not a universal correction for mismatched Reynolds number. State what was matched and measure the residual sensitivity.

Mach number and other ratios

Mach number $M = U/a$ compares speed with the speed of sound. With $a \approx 343$ m/s at about 20°C, 100 km/h is $M \approx 0.081$ and 300 km/h is $M \approx 0.243$. Incompressible modeling is often a reasonable starting point at ordinary car speeds, but evaluate local acceleration and the required accuracy rather than treating 0.3 as a hard physics switch. The sound speed and density both vary with atmospheric conditions.

Dimensionless geometry is equally important: wing height/chord, floor clearance/length, diffuser exit/throat area, wheel diameter/length, and splitter overhang/clearance. Strouhal number $St = fL/U$ is useful when comparing wake frequencies. A solver's pseudo-time iteration history is not automatically a physical frequency record. You need a time-accurate method and adequate sampling before inferring an unsteady loading frequency.

Worked example

At 100 km/h with $A = 2.2$ m², $C_d = 0.32$, $C_l = -0.40$, and $\rho = 1.225$, $q = 472.61$ Pa, $D \approx 332.7$ N, downforce ≈ 415.9 N, and still-air aero power ≈ 9.24 kW. At 200 km/h the forces are four times larger and the power is eight times larger if the coefficients and geometry stay constant.

▼ **Check: car A has $C_d = 0.30$ with $A = 2.5$ m²; car B has $C_d = 0.33$ with $A = 2.1$ m². Which has less drag at the same speed?**

Car B: its drag area is 0.693 m² versus 0.750 m² for car A. C_d alone would give the wrong ranking.

Definition reference: NASA dynamic pressure. The workbench and scaling examples are independently calculated.

DAY 1 / 04 / FOUNDATIONS

Boundary layers and separation

You will recognise pressure recovery, understand why flow can detach, and distinguish turbulence from separation.

The slow air near a wall changes everything

For an ordinary stationary solid wall, the fluid velocity at the wall equals the wall velocity. In the car-fixed frame the body is stationary, while the road and tyre surface move. Away from the wall the flow approaches an external velocity. Between them lies a boundary layer with strong velocity gradients and viscous shear. It grows along the surface and changes the effective displacement of the outside flow. A boundary layer is not a uniform film of fixed thickness.

A laminar boundary layer has ordered mean motion with relatively weak mixing across its thickness. A turbulent one has velocity fluctuations and stronger momentum transport. A turbulent profile is often fuller and can tolerate stronger pressure recovery before separating, but it generally has more skin friction.

Transition does not equal separation; turbulence can exist in an attached layer. Likewise, low turbulence in a plot does not establish that a flow is attached or efficient.

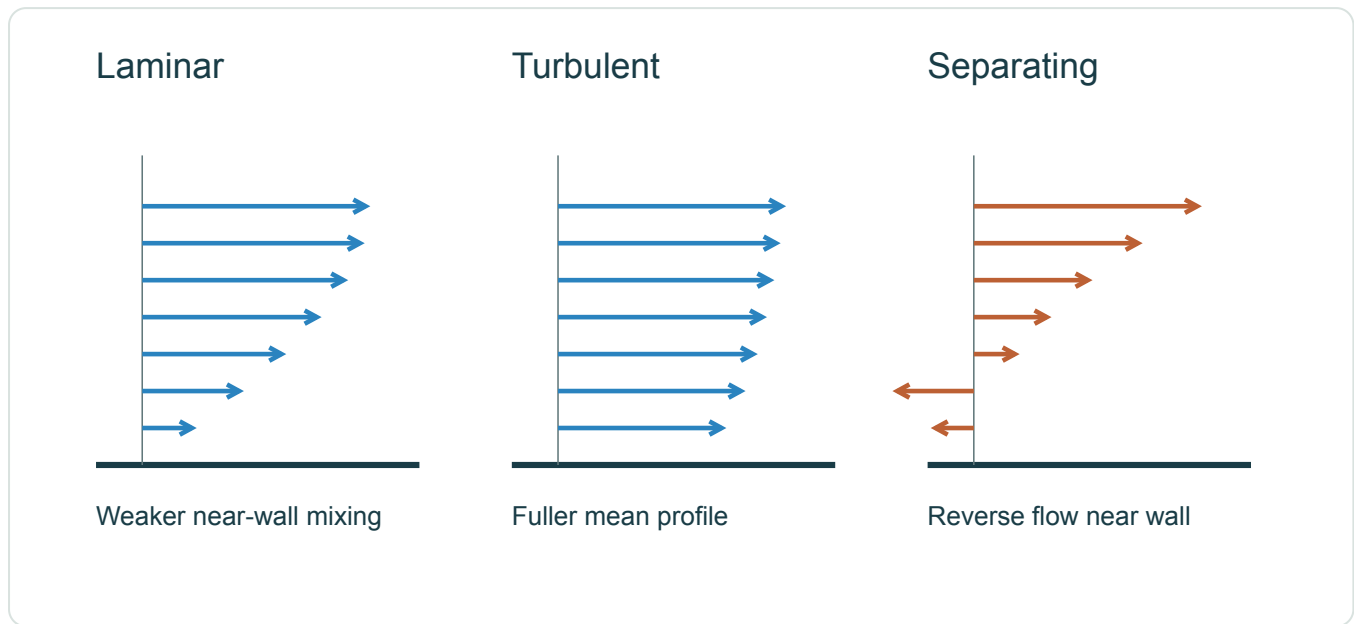


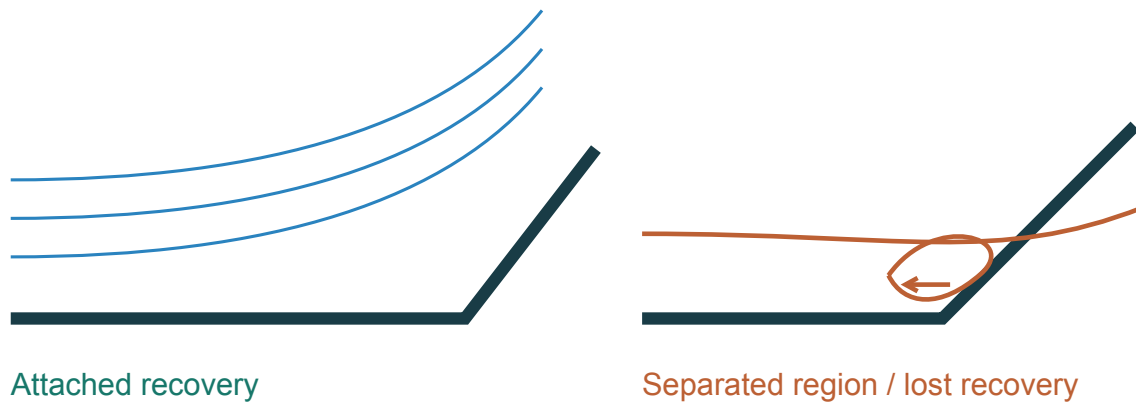
Figure 5. Illustrative velocity profiles for laminar, turbulent, and separating flow. Horizontal arrows denote velocity relative to the wall.

Pressure recovery is an uphill climb

When pressure decreases downstream, the pressure gradient accelerates the external flow. This is a favourable gradient. When pressure increases downstream, it decelerates the flow and is adverse to the near-wall layer. The slowest fluid may lack enough momentum to negotiate that pressure rise. Wall shear approaches zero; reverse flow can develop, and the layer detaches. The exact separation behaviour depends on upstream history, geometry, three-dimensionality, and unsteadiness.

A diffuser intentionally recovers pressure by expanding a passage. A rear body taper also asks air to recover pressure after the roof or sides. Both can succeed when the recovery is gradual enough for the incoming layer, and fail when it is too demanding. Simply making a ramp steeper creates more expansion, but can also create a detached region that blocks effective flow. A longer smooth recovery may cost packaging space or wetted-area friction. This is the origin of a design tradeoff, not a universal “best angle.”

Pressure recovery asks slow near-wall air to decelerate



Angle, upstream layer, leakage, and ground proximity all matter.

Figure 6. Attached and separated pressure recovery on a ramp. The paths are explanatory drawings; neither is a numerical prediction.

Learn the diagnostic combination

Look for local skin-friction direction, reversed surface flow, near-wall velocity profiles, recirculation, a pressure plateau, and the growth of a wake. A single centreline streamline can miss separation near a diffuser side edge. A mean streamline can conceal intermittent detachment. On a three-dimensional surface, a separation line is not fully described by the simple two-dimensional rule “the tangential velocity gradient is zero.” In practice, surface-flow topology and off-wall sections belong together.

If the app has only a coarse wall model, a colourful surface-friction map may not resolve the actual near-wall physics. Ask whether the grid, wall distance, and turbulence treatment support the feature you are interpreting. A 5 mm Gurney flap or 10 mm tyre-floor gap cannot be judged from a 60 mm field sample just because the geometry viewer draws it sharply.

Useful flat-plate estimates, with narrow scope

For an ideal zero-pressure-gradient flat plate, local laminar estimates include $\delta/x \approx 5/\sqrt{Re_x}$ and $Cf_x \approx 0.664/\sqrt{Re_x}$. A common smooth turbulent local estimate is $Cf_x \approx 0.0592 Re_x^{-1/5}$ under its fully turbulent assumptions. These are not whole-car drag formulas, not reliable separation criteria, and not a transition model. Use them to get an order of magnitude, then ask why a roof, rotating tyre, or diffuser violates the assumptions.

Worked example

At $U = 30$ m/s and $x = 1$ m, $Re_x = 2$ million. The ideal laminar thickness estimate is 3.5 mm. A rough turbulent thickness estimate $\delta/x \approx 0.37 Re_x^{-1/5}$ gives about 20 mm. Neither value accounts for the car's pressure gradient or upstream disturbances; their difference explains why boundary-layer state matters for a small gap.

▼ **Check: should you make every surface as laminar as possible to reduce drag?**

No. Laminar flow can reduce friction but may separate under pressure recovery that an attached turbulent layer would tolerate. You must evaluate the combined friction and pressure-drag change.

Introductory context: NASA boundary layers. The estimates here are standard limited models, not car-specific correlations.

DAY 1 / 05 / FOUNDATIONS

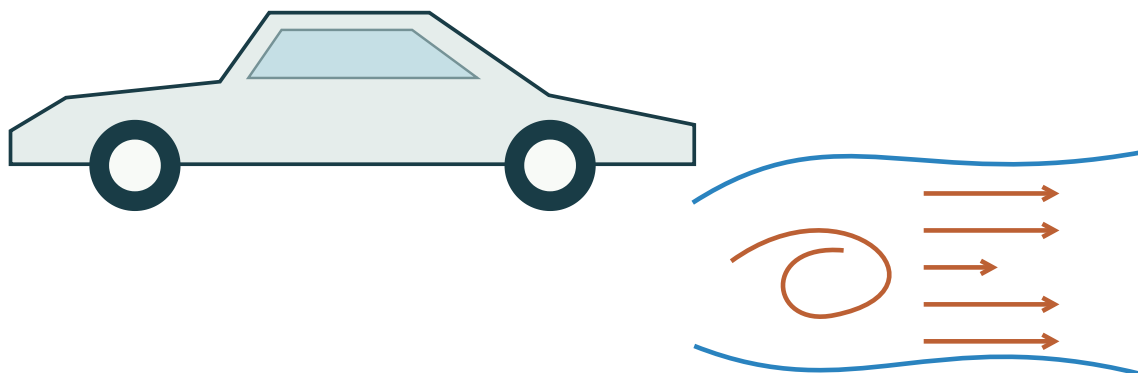
Where drag comes from

You will identify plausible drag sources and understand why smoothing a surface may not reduce whole-car drag.

Pressure drag and friction drag

Pressure produces drag when its normal force has a component opposing the car's motion. A blunt front creates relatively high pressure, while the rear wake can leave low base pressure. The incomplete rear recovery creates a large pressure-drag contribution. Skin friction is the streamwise component of tangential stress over the wetted surface. Their relative importance depends on the geometry and conditions. Do not import a fixed percentage from another vehicle to decide what matters on yours.

Roof, sides, underbody, wheels, and cooling passages all influence the wake. A smoother floor may reduce frictional disturbance and improve pressure recovery, but it may also increase flow through a constricted passage, alter lift, or change wheel interactions. A carefully placed spoiler can fix a separation location and sometimes improve base pressure; another one can lower base pressure and increase drag. A clean-looking smoke pattern is a hypothesis about the mechanism, not a measurement of the total drag.



Incomplete rear pressure recovery

A single near-wake slice is not the complete drag balance.

Momentum / total-pressure deficit

Figure 7. A schematic wake has a momentum deficit and loss of total pressure. Static-pressure colour alone is not a drag budget.

A wake is more than a trail of vortices

The flow behind a car can contain recirculation, shear layers, streamwise vortices, and asymmetric states. Vorticity is local fluid rotation; a coherent vortex is a particular organised structure, not every region with nonzero vorticity. Vortices can help keep a flow attached or isolate a floor from side leakage. They also require kinetic energy and can strengthen mixing losses. “Remove all vortices” and “add a vortex” are both incomplete design instructions.

In a simple steady far-wake momentum analysis, drag relates to the momentum deficit integrated over a suitably chosen control surface, with pressure, transverse flow, and other terms treated consistently. Looking at one velocity slice does not supply all those terms. Near the car, pressure gradients and lateral velocities are especially important. A wake image is excellent for locating candidate problems but cannot replace a force integration.

Lift-related drag is a separate physical cost

A finite-span lifting device generates induced flow. Its aerodynamic force is tilted and has a drag component associated with producing lift or downforce. For an isolated attached wing, lifting-line models give $C_{di} \approx C_l^2 / (\pi e AR)$. The square means downward lift has the same induced-drag sign as upward lift. This model does not capture all profile drag, supports, interactions, or ground effect. A diffuser can also affect the energy and momentum of the wake; it is not exempt from an energy cost simply because it is hidden beneath the car.

Changing frontal area, wheel exposure, mirror shape, cooling inlet/outlet placement, underfloor roughness, or rear taper may offer better drag reduction than reducing wing angle. Start by measuring the whole-car force and locating which flow regions change. A large drag contribution on a part is not necessarily removable without affecting the rest of the car. Delete a mirror in a simulation and you change more than its standalone projected area.

The right comparison controls

Use identical speed, density, reporting area, ride height, yaw, road motion, wheel treatment, domain, and numerical effort. Save shared colour scales and the same camera. Compare the force history before choosing a favourite flow image. If the baseline has not settled and the variant has, their final numbers can reflect sampling differences. Small drag gains need more scrutiny than spectacular-looking pictures.

Lab 02 · 45 min · Read the baseline

Open the teaching pack from the course workbench in Lesson 17. Keep only Flat-floor body and Wheels active. Record dimensions and the 10 mm wheel-road gap, confirm geometry, and run at 100 km/h, zero yaw, moving road, rotating wheels, $A_{ref} = 2.2 \text{ m}^2$, and $p = 1.225$. Use Custom 72 cells per car length and 10 passes, detail cells off, to match the included examples. First inspect the history, then pressure, vertical streamlines, surface flow, and wake losses.

Predict: mark likely high-pressure and low-pressure regions before viewing them. **Observe:** save two views and the signed lift. **Interpret:** connect one pressure region to its force direction and one separation hypothesis to a diagnostic you would need. If the run is unsettled, state that explicitly and use its history rather than its final sample.

Background: NASA's overview of drag. The diagnostic workflow refers to the current EasyCFD interface.

DAY 1 / 06 / FOUNDATIONS

How to read a flow visualisation

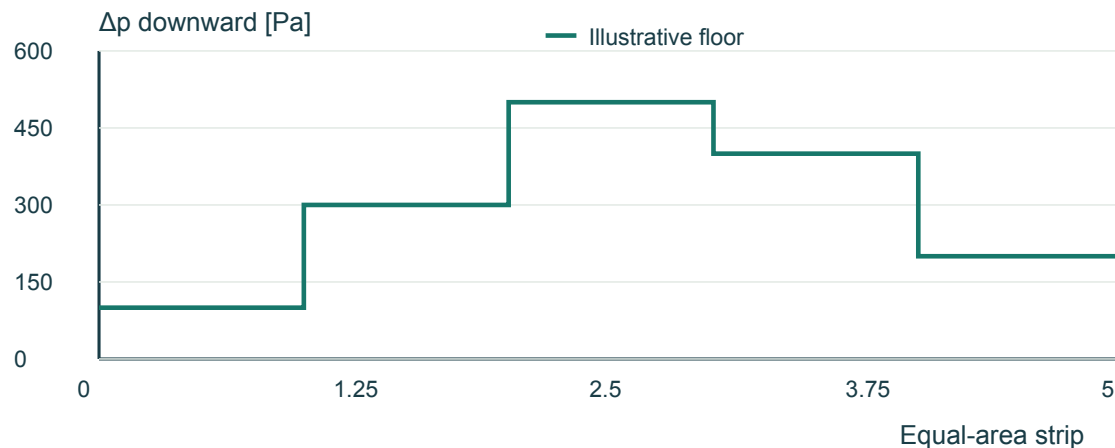
You will choose a view that answers a question and avoid common mistakes with colours, smoke, and histories.

Pressure: always ask about the normal

A pressure map tells you how local pressure differs across the surface. It does not directly display downforce. On horizontal top and bottom faces the force directions are opposite. On a steep diffuser ramp, pressure force has both vertical and streamwise components. A large negative- C_p area may create suction with a substantial drag component, depending on its slope. For a device comparison, use the same C_p range and inspect the whole body as well as the device.

A C_p colour is nondimensional; a pascal colour depends on density and speed. Increasing speed can intensify a pressure-in-pascals plot even if the coefficients stay the same. Independent autoscaling can make a weak and strong pressure field look identical. A narrow extreme on a small area can dominate a legend while contributing little to the integrated load. Note whether a plotted field is a final snapshot, an average, or a reconstruction of averaged quantities.

A pressure difference needs area and surface direction



5 × 0.20 m² strips → 300 N total; a tiny extreme need not dominate load.

Figure 8. Integrating a simplified pressure difference over equal-area floor strips. The curve and values are invented for this teaching calculation, not CFD output.

Lines are not all the same

A streamline is tangent to the velocity field at an instant or to a chosen mean field. A pathline traces one particle through time. A streakline connects particles released from a source over time. They coincide in steady flow, but not generally in unsteady flow. EasyCFD's steady-field streamlines and smoke are primarily

visualisations of its saved field. Solver iteration evolution and locally stepped pseudo-time are not a time-accurate physical film of your car's flow.

Move the seed rake. A few lines above a wing can miss its underside and tip vortices. Seeds inside solids, near poorly sampled cells, or in a clipped plane can disappear without proving a physical blockage. Inspect several heights and lateral positions. Read the legend and the camera's coordinate frame. Use vertical planes for roof and floor attachment, horizontal planes for side leakage, and cross-sections for wheel wakes and diffuser outflow.

Total pressure, turbulent kinetic energy, and surface flow

A low-total-pressure wake indicates loss of useful mechanical energy relative to the freestream, whereas a low static pressure can be part of efficient reversible acceleration. A turbulent-kinetic-energy map describes the modeled or resolved fluctuation scale appropriate to the solver, not a universal "bad flow" indicator. The $k-\omega$ SST model predicts statistics of turbulence; it does not show every real eddy. Surface flow and wall stress can suggest attachment and crossflow, but only to the fidelity supported by wall sampling and geometry resolution.

Force histories deserve the same care as flow images. A steadily falling residual measures how well the discretised equations are being solved; it does not establish that the chosen model or mesh matches reality. A flat C_d trace can coexist with changing C_l or pitching moment. Report the averaging window and compare its two halves. When an oscillation remains, extend the sample and quantify the mean's stability. The EasyCFD $\pm$ band is a spread of sub-window means, not a statistical 95% confidence interval and not total aerodynamic uncertainty.

Worked example

Five floor strips each have 0.20 m^2 area and mean downward pressure differences of 100, 300, 500, 400, and 200 Pa. The approximate floor load is $0.20 \times 1500 = 300 \text{ N}$. A 900 Pa spot over only 0.005 m^2 contributes 4.5 N. Areas and normals matter more than the brightest pixel.

Lab 03 · 45 min · Test a visual explanation

Take your baseline from Lab 02 and write "I think the rear wake is associated with incomplete pressure recovery." Save a roof/centreline velocity section and a rear pressure view. Move a horizontal section across the wheels and floor. Find evidence that supports the statement and one feature the first view missed. Export the result JSON and force-history CSV from the app. In your notebook record the averaging window, signs, legend scales, and whether the run settled.

Debrief: explain why a fast blue region, a low-pressure blue region, and a low-total-pressure blue region can mean three different things. If you cannot identify the plotted variable and its units, you cannot use that image as engineering evidence.

▼ **Day 1 checkpoint: your Cd changed from 0.34 to 0.32 after you changed Aref. Did you improve the car?**

No aerodynamic change has been established. A different denominator changes the reported coefficient. Compare dimensional forces or convert both to a common area, then check all physical conditions.

Wings: section shape, incidence, and stall

You will distinguish section geometry from effective incidence and recognise the limits of a simple wing calculation.

An inverted airfoil is a pressure device

A race-car wing is shaped and oriented to create lower pressure mainly on its lower surface and higher pressure mainly on its upper surface. Camber is the curvature of its mean line; thickness describes its envelope. Chord is the leading-edge-to-trailing-edge distance and span is the lateral extent. A symmetric section can make downforce at incidence; camber shifts its zero-lift angle. The integrated difference between upper and lower pressure is the main lift contribution for an ordinary attached wing, while shear and pressure recovery contribute drag.

The nominal mounting angle is measured relative to a reference on the car. The effective angle is measured relative to the local incoming air. A wing behind a sloping rear screen may see inclined, slower, turbulent air. Changing wing height can change local speed, incidence, and wake interaction at once. If local speed is 80% of freestream, local dynamic pressure is 64% of freestream. A coefficient derived in clean air at the freestream speed will overstate the load if applied unchanged to that slower flow.

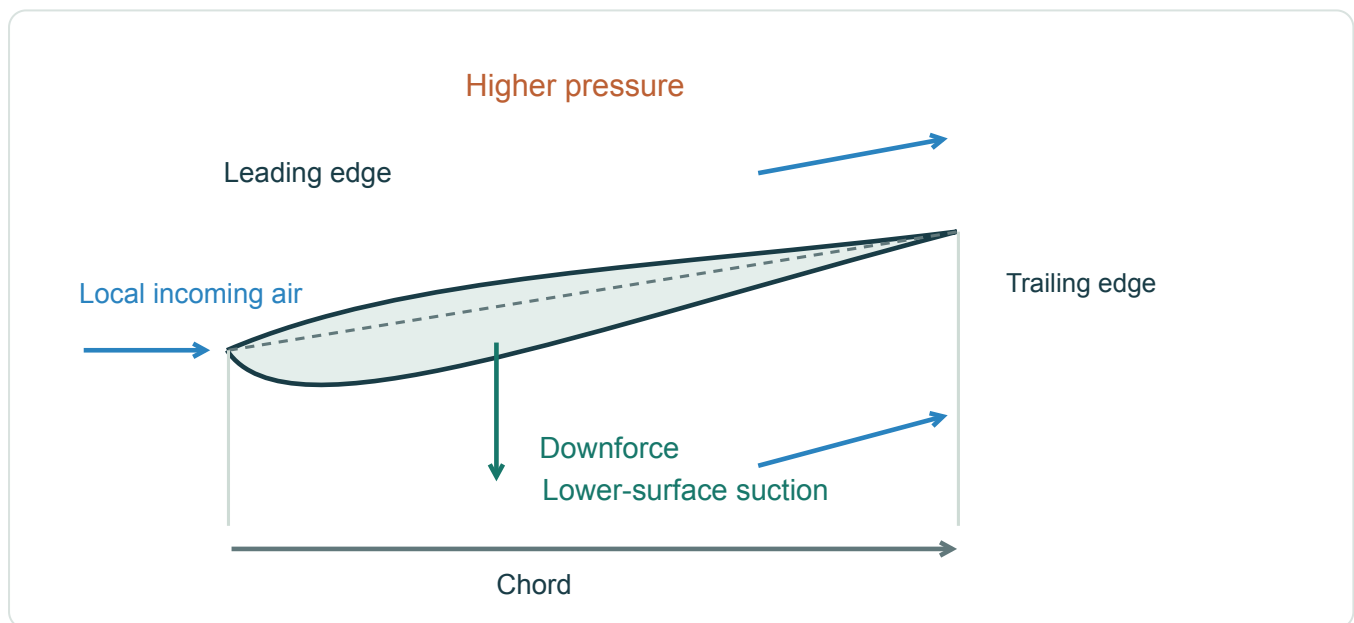


Figure 9. Section terms and local incidence on an inverted wing. The lower-surface suction and upward air turning are schematic.

Start with a linear model, then locate its boundary

For an ideal thin two-dimensional airfoil in attached incompressible flow, the lift slope is approximately 2π per radian relative to zero-lift incidence. A finite wing's slope is smaller because induced flow changes the effective angle. In the workbook below, define a positive downforce incidence α and an assumed zero-downforce incidence α_0 . It estimates the magnitude coefficient $C_N = a(\alpha - \alpha_0)$, then profile and induced drag. This deliberately omits separation, body interference, endplates, ground effect, transition, and supports.

$$a \approx 2\pi / [1 + 2/(e AR)] \text{ per radian}$$

$$C_{N,\text{wing}} \approx a(\alpha - \alpha_0) \quad ; \quad C_{D,\text{wing}} \approx C_{D0} + C_{N,\text{wing}}^2 / (\pi e AR)$$

Workbench · Finite wing tradeoff

Teaching model: attached isolated wing, prescribed $e = 0.8$, $C_{D0} = 0.02$, and $\alpha_0 = -2^\circ$. These assumed parameters are not measured polars for the supplied NACA geometry. High incidence remains an extrapolation, never a stall prediction.

Downforce incidence	6°	Span	1.7 m
Chord	0.32 m	Freestream speed	100 km/h
Local / freestream speed	1		

0.544 m²

Wing planform area

5.31

Aspect ratio

0.597

Wing CN (planform area)

153.4 N

Teaching-model downforce

12.0 N

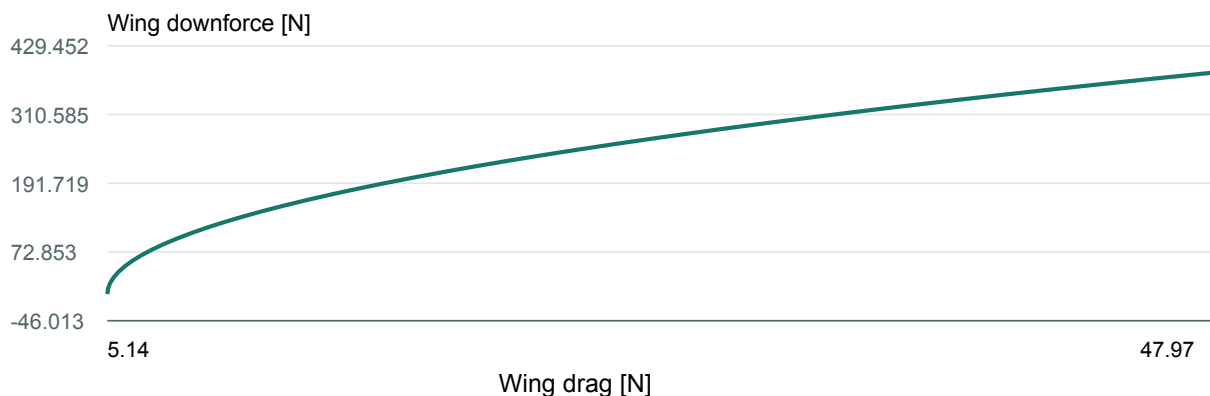
Profile + induced drag

0.027

Induced Cd only

Attached-flow estimate only. The installed NACA geometry may have a different polar and inflow.

Assumed attached-wing force tradeoff



Stall is a flow-state change

Increasing incidence makes the pressure recovery on the suction surface more demanding. Separation can grow and reduce lift slope, produce a plateau, or cause a sharp loss of downforce. Drag usually increases and the pressure centre may move. There is no universal automotive stall angle. Reynolds number, leading-edge geometry, roughness, upstream turbulence, ground proximity, and finite-span interactions all matter. A smoothly rising linear calculator cannot tell you when your installed wing stalls.

A multi-element wing uses gaps and carefully arranged elements to make stronger total turning while managing the boundary layers and pressure recovery of each element. The gap and overlap are aerodynamic parameters. Arbitrarily stacking two wings does not reproduce a designed slot. If the gap is unresolved by the mesh, a solver may effectively fuse or block it. The additional downforce can come with more drag and sensitivity, so compare the complete package through its actual operating range.

A practical first study

Keep section, span, chord, endplates, position, and supports fixed while testing a small sequence of mounting incidences. Inspect the incoming flow ahead of the wing, pressure recovery on the lower surface, spanwise load, and the whole-car moment. If downforce stops rising, investigate attachment and convergence before assuming a physical stall. Repeat at another mesh resolution and compare against a suitable experiment or qualified solver. EasyCFD's documented browser-wing ranking limitations make this last step especially important.

Lab 04 · 60 min · Two wing incidences

On the teaching car, leave Flat-floor body and Wheels active; enable Rear wing 6 deg, run, then disable that group and enable Rear wing 12 deg. Keep the parts' own switches on so both versions still shape the common grid. Compare total force, the wing's force, and the body change. Inspect the local inflow and lower-surface flow at several span positions. Use the saved course runs while repeats compute.

Predict: the attached-flow model's direction of change. **Interpret:** if the CFD ranking disagrees, list numerical resolution, flow-state, and inflow explanations. The learning outcome is a defensible next test, not a verdict that 6° or 12° is the better real wing.

Model reference: NASA induced-drag coefficient. Ground-proximity background: Zeriha's experimental wing-in-ground-effect thesis.

DAY 2 / 08 / DEVICES

Finite span, endplates, Gurneys, and mounts

You will recognise three-dimensional wing costs and treat the installation as part of the device.

The span changes the price of downforce

For a rectangular wing, planform area $S = bc$ and aspect ratio $AR = b^2/S = b/c$. At the same wing coefficient, an isolated wing's induced-drag coefficient decreases with increasing aspect ratio. At a fixed required dimensional load, ideal induced drag can be written $D_i = N^2/(\pi e q b^2)$. This reveals why span matters: for

fixed load, q , and efficiency factor, more span reduces the ideal induced penalty. It does not mean an arbitrarily wider wing is feasible or better in the installed car's flow.

Width restrictions, visibility, structural stiffness, roof wakes, yaw, and the tyre track create practical limits. Spanwise loading near a wheel wake may be worse than clean-air theory predicts. A wing may have nearly uniform section geometry but a very nonuniform pressure distribution. Look at several sections and integrated spanwise load, not just the centreline. An efficiency factor e is a modeling summary of loading, not a universal constant you can read from a wing's appearance.

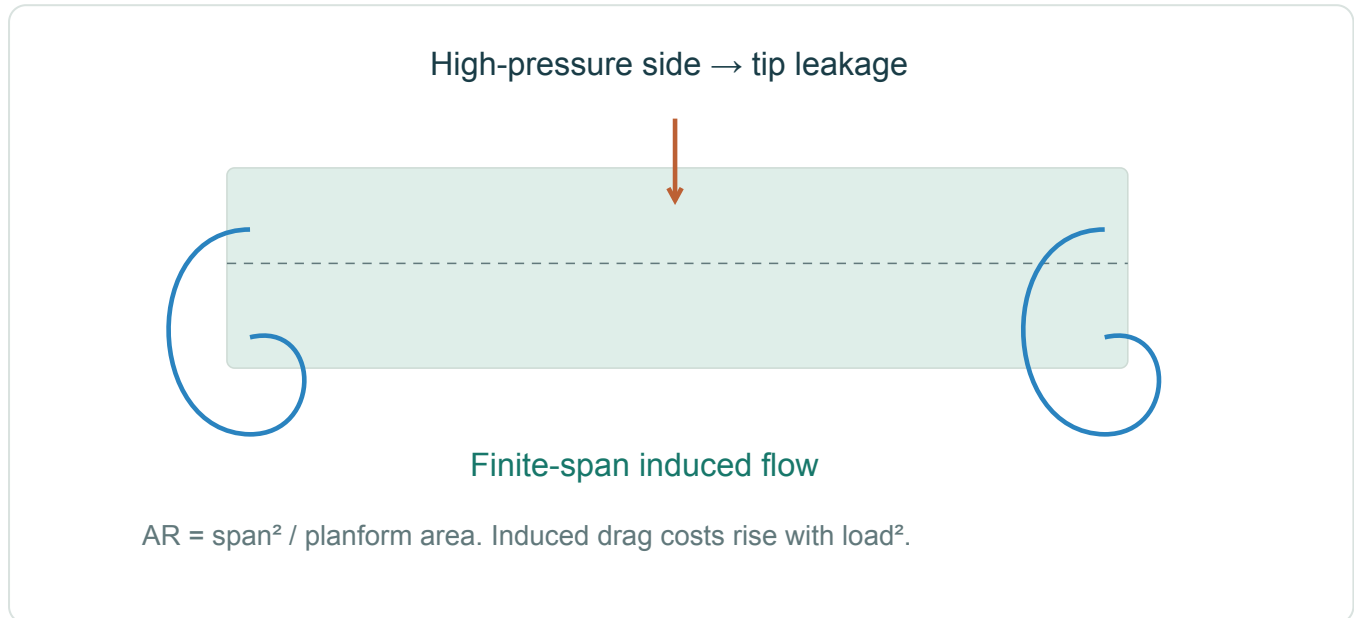


Figure 10. Finite-span pressure leakage and tip vortices on a downforce wing. The vortex sense is shown schematically, not as a CFD result.

Endplates constrain leakage; they also have drag

An endplate can impede flow from the high-pressure side to the low-pressure side around the tip, alter the loading, and help manage tip flow. It is not an infinite-span wall. Its size, overlap with the wing, proximity to body surfaces, shape, and yaw all matter. Extra wetted area and exposed frontal area contribute drag, and in crosswind it becomes a side-force surface with a moment arm. Compare the wing plus endplates plus supports, including changes to the car wake.

A Gurney flap is a small tab near the trailing edge, typically on the pressure side of a downforce wing. It can increase turning and change the rear pressure field, raising load and commonly increasing drag. The relevant size is h/c , not merely millimetres. A 5 mm tab on a 250 mm chord is 2% chord. Whether that is useful depends on the section and conditions. A coarse zero-thickness representation may not capture the small recirculating structure behind the tab.

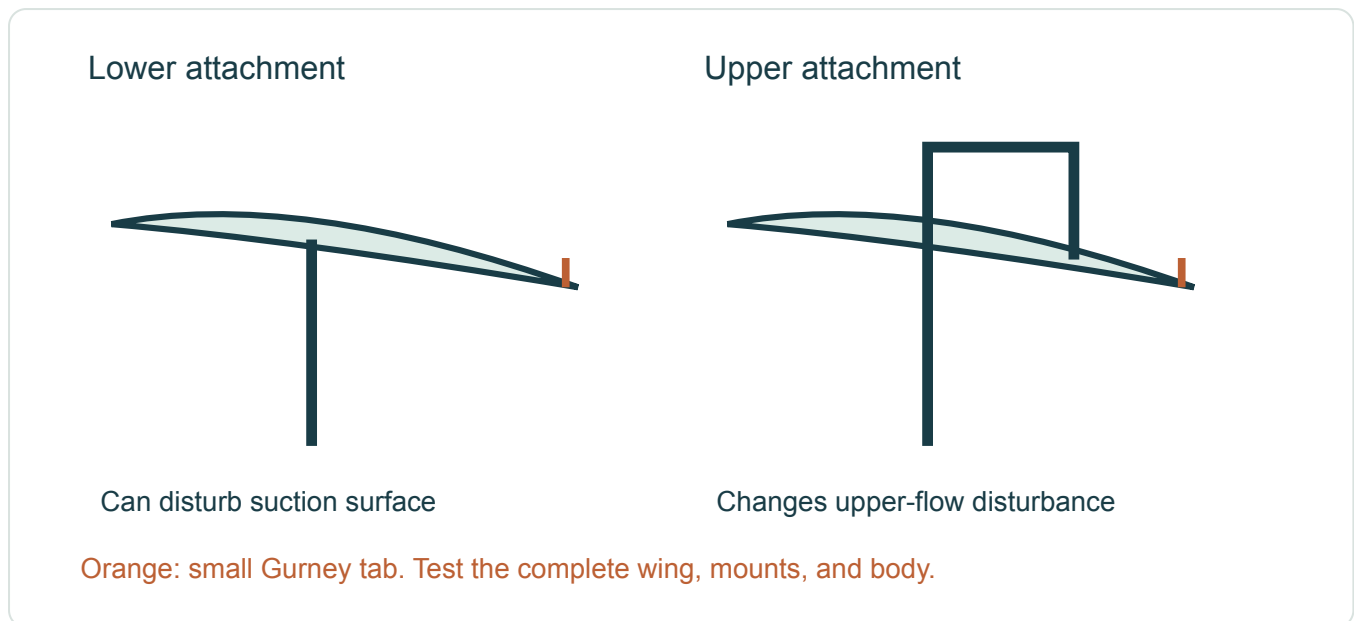


Figure 11. Conventional and upper mounts, a trailing-edge tab, and local wake interaction. This drawing explains questions to test; it does not establish an installation winner.

Supports and real load paths

Supports that disturb a suction surface can degrade the wing's useful pressure recovery. Upper “swan-neck” arrangements move some attachments away from the lower suction surface, but their own shape, stiffness, and location still affect drag and loading. A mount in clean upper flow may create a different penalty from one in a body wake. Avoid treating a style of support as automatically superior without a controlled installed comparison.

The aerodynamic force must reach the chassis through a credible load path. A 300 N wing result at 100 km/h becomes about 1200 N at 200 km/h under constant coefficients, before transient factors and structural margins. Mounting deflection changes incidence and can alter the aero load itself. A flexible boot lid may change both geometry and load distribution. The teaching models omit wing mounts intentionally; that omission makes them suitable for mechanism exercises, not construction qualification.

Position and moment can dominate the result

A wing behind the rear axle adds rear load and can unload the front through its moment. Raising the wing can change inflow and the moment produced by drag acting above the road. A taller wing with more total downforce can therefore make the car's front share worse. Study load location, vertical height, and pitch moment as well as total force. Vehicle balance is covered rigorously in Day 4.

Worked example

For $N = 500$ N, $q = 600$ Pa, $b = 1.6$ m, and $e = 0.8$, ideal induced drag is 64.8 N. At $b = 1.9$ m it is 45.9 N if load and e stay fixed. This 18.9 N saving excludes profile drag, endplates, supports, and changes in local inflow. Do not present it as a measured whole-car gain.

▼ **Check: does reducing chord always improve an installed wing because AR rises?**

No. At fixed span and required load, ideal induced drag depends on span, while changing chord alters section coefficient, Reynolds number, profile drag, stall margin, and structure. AR alone is not an installed design decision.

DAY 2 / 09 / DEVICES

Splitters, air dams, and front load

You will explain a splitter's pressure difference and distinguish its function from an air dam and a decorative lip.

The front flow has choices

At the nose, approaching air divides over, under, and around the car and through open cooling passages. A splitter is a roughly horizontal surface extending near the lower front edge. High pressure on its upper side and lower pressure beneath it can generate downward load. Its leading edge, connection to the floor, side leakage, ground clearance, and the nearby wheel flow determine the actual distribution. The front fascia's stagnation region and cooling inlet cannot be treated as separate from this flow split.

An air dam is primarily a vertical barrier that reduces or redirects underbody inflow. It can reduce a messy underbody's drag or change lift, but also adds pressure drag itself. A splitter deliberately offers horizontal area on which a pressure difference can act. A lip may combine both effects depending on shape. Product names do not define the physics; inspect the surfaces, the flow path, and the resulting pressures.

Horizontal splitter ≠ vertical air dam

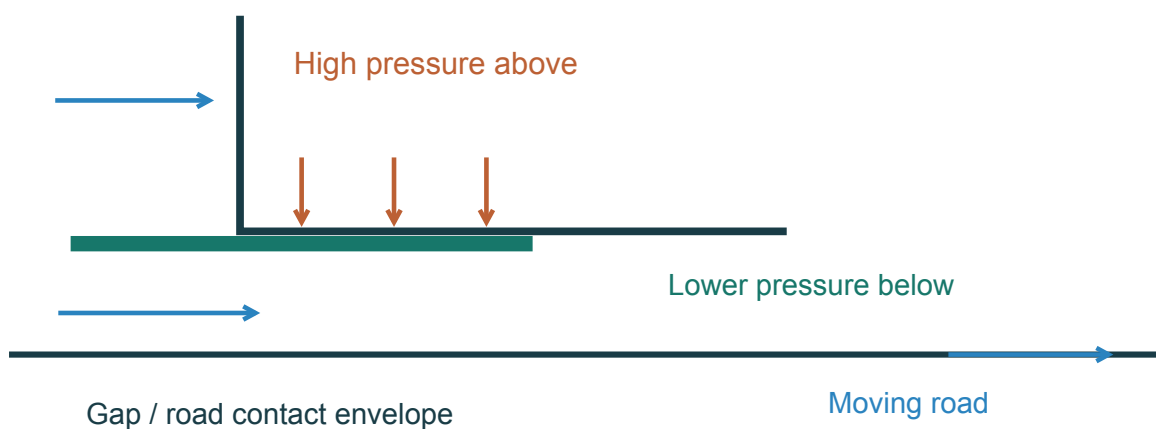


Figure 12. Splitter and air-dam mechanisms, with side leakage and a moving road. Pressure arrows are schematic.

A sensible first load estimate

If a 0.35 m^2 splitter experiences a mean pressure difference of 350 Pa downward, the estimated load is 122.5 N . This is a bookkeeping example, not a way to prescribe the pressure difference. Increasing overhang increases available area but can change the field, front-wheel interaction, bending, and ground-contact risk. You cannot assume load rises linearly with overhang once the flow or geometry changes.

Clearance can be crucial. A smaller gap may increase acceleration and suction, but it can also reduce mass flow, increase leakage sensitivity, or create intermittent blockage. Braking pitch, bumps, kerbs, and aerodynamic suspension compression all reduce the available gap. A rigid static simulation with a perfectly smooth road misses those operating conditions. State the loaded clearance, not just the dimension measured while parked.

The floor connection matters

A splitter ending at a rough open underside may not feed a stable floor flow. A smooth tray can reduce disturbance and interact with the rear diffuser. Sealing the top side against high-pressure leakage changes the pressure difference; adding a side fence changes three-dimensional flow. A front modification can starve a rear device or improve its inflow, so whole-car force and axle balance must be measured. The drag of the splitter group alone cannot represent that system change.

A splitter ahead of the front axle has a large front-loading moment arm. That may be useful, but the rear equivalent load can decrease for a given point load. Also account for drag at its height and any rear aerodynamic changes. Its load path may pass through bumper skins, brackets, and chassis mounts. The pressure distribution affects bending: two splitters with the same total force can have different peak bending moments and support needs.

Test a mechanism, not a photograph

For a first study keep width, leading-edge shape, floor connection, and position fixed. Compare a baseline and one overhang. Examine upper/lower C_p , front-wheel cross-sections, floor mass-flow behaviour, and axle load. Then sweep clearance. If the geometry looks good but the computed splitter force is negligible, determine whether the app represented its thickness and leading edge adequately before discarding the concept.

Lab 05 · 60 min · Splitter and canards as front devices

Run the flat-floor baseline, splitter only, canards only, then splitter plus canards on the same grid. Keep wings, diffuser bodies, and skirts off. Record total ΔD and ΔN , group forces, and the body/wheel change. Sample surface pressure above and below the splitter and horizontal flow near the front wheels.

Deliverable: a four-row table and a statement about interaction: is the combined change approximately the sum of the separate changes, within the observed numerical variability? A small unexplained discrepancy is not proof of synergy. Canard mechanisms are developed in the next lesson.

▼ **Check: why can a small splitter outperform a large one in a real package?**

Its pressure difference, leakage control, clearance, attachment, or inflow may be better, and it may retain useful front load over more operating conditions. Size alone does not establish integrated or robust performance.

DAY 2 / 10 / DEVICES

Canards, vortices, and wheel-flow control

You will separate a small device's direct force from its effect on downstream surfaces.

Small surfaces can change a large flow system

Automotive canards, or dive planes, are small inclined surfaces near the front corners. They can contribute direct downforce and generate vortical flow that changes side leakage, wheel wakes, or downstream attachment. A modest direct device load can accompany a larger body change. The reverse also occurs: a useful device force can be outweighed by a detrimental downstream change. Evaluate the difference on the whole car and retain the group breakdown as an explanation.

Pressure tends to equalise around a free edge, causing spanwise motion and a rolled-up vortex. A vortex may draw higher-momentum air into a boundary layer or create a pressure barrier to leakage. These are hypotheses about a specific installation. The vortex trajectory depends on local flow, yaw, position, other vortices, and the wheel. A hand-drawn line from a canard directly to a diffuser is not evidence that the real vortex reaches that location intact.

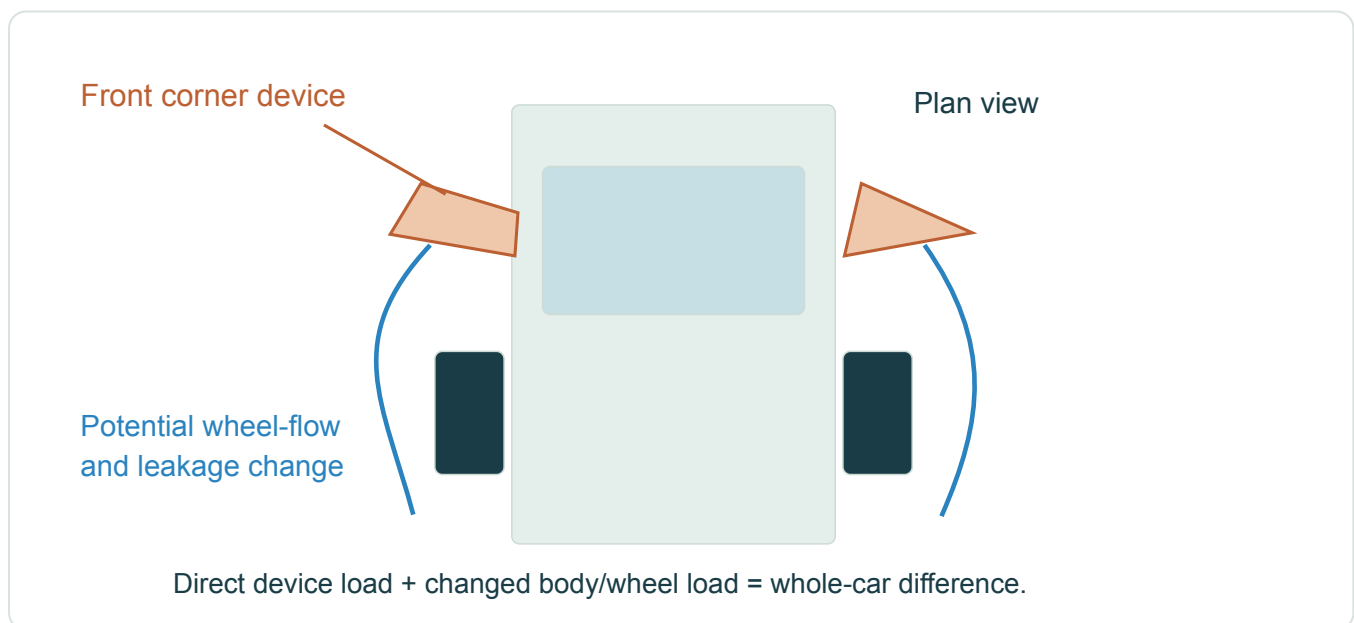


Figure 13. Direct canard force and a possible downstream flow influence. The proposed vortex route is schematic and requires testing.

The rotating wheel is an active aerodynamic obstacle

A wheel rotates while translating with the car. In a car-fixed model its centre is fixed and its surface velocity varies around the circumference. At the nominal road contact the surface velocity approaches the road's velocity. Tyre shoulder shape, wheelhouse openings, steering, rim ventilation, contact-patch treatment, and the road gap all influence the wake. A smooth cylinder is a simplified teaching model, not a real tyre.

Wheel deflectors can redirect approaching flow, but a change in front-wheel flow can alter floor suction and rear-wheel conditions. Covering a wheel can reduce one drag source while changing cooling or force balance. A coarse representation of the contact patch may create an artificial jet through a gap or excessive blockage. The supplied pack uses cylindrical wheels with a deliberate 10 mm road gap to suit the app's geometric assumptions. That gap is a numerical exercise condition, not a recommendation for a physical tyre.

How to observe a vortex responsibly

Use cross-sections at several downstream stations with velocity vectors or appropriate vortex diagnostics, and track surface pressure and integrated force changes. Turbulent kinetic energy alone is not a vortex detector. A single vorticity-magnitude contour can light up a shear layer as well as a coherent vortex. If using Q-criterion or another diagnostic, record its threshold and sign convention. Test whether the conclusion survives a changed threshold, a finer mesh, and small yaw.

Control canard angle, area, position, and count separately at first. Adding several surfaces at once prevents attribution. Because these devices are small, geometry resolution and wall modeling are critical. A canard represented by very few cells may still appear smooth in the renderer. Real improvement must exceed the numerical sensitivity and be confirmed in relevant conditions. The current browser solver is especially exploratory for small downforce devices.

When the direct-device result misleads

Suppose canards produce 40 N direct downforce and 12 N drag. The body/floor change adds 25 N downforce and removes 5 N drag; total improvement is 65 N downforce for 7 N added drag. Another installation may give the same direct 40 N but unload the floor by 60 N; then the package loses 20 N downforce. This is why a device-only efficiency rating cannot decide a complete car modification.

▼ Check: a canard's downstream smoke trail is visible. Does that prove it seals the underfloor?

No. Trace the relevant pressure and crossflow changes and demonstrate the whole-car force effect. A visible trail does not establish leakage suppression or useful load.

DAY 2 / 11 / DEVICES

Floors and diffusers: acceleration and recovery

You will explain the complete underfloor path and distinguish ideal recovery from a prediction of downforce.

The diffuser is the rear part of a connected passage

A low floor passage can accelerate air and establish low static pressure beneath the car. A rear diffuser increases the passage height toward the exit, allowing the flow to decelerate and recover pressure. Successful recovery helps the system sustain useful underfloor flow and suction upstream. The downward load commonly comes from a broad floor area as well as the ramp. Calling the diffuser a device that “sucks air out” skips the pressure and mass-flow solution that you need to understand.

The upstream entry, throat height, ramp length, side leakage, wheel wakes, exit location, and rear wake all contribute. A ramp bolted under an obstructed suspension is not equivalent to a coherent floor feeding the same ramp. The exit does not necessarily see freestream static pressure. A rear wing can change the pressure above and behind the diffuser, altering its flow. That is a possible interaction to test, not guaranteed free downforce.

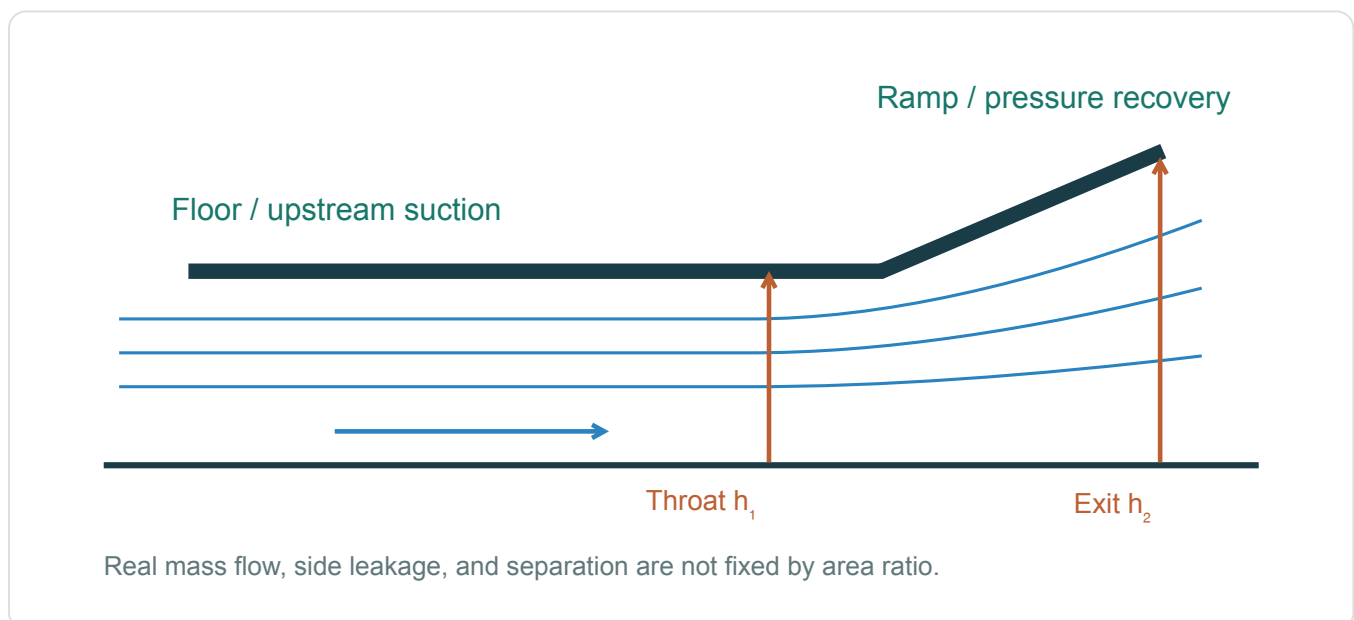


Figure 14. Underfloor entry, throat, ramp, and recovery. Pressure trend is a schematic; real flow is three-dimensional and lossy.

A limited one-dimensional recovery model

For a constant-width passage with throat height h_1 and exit height $h_2 = h_1 + \ell \tan \theta$, the geometric area ratio $R = h_2/h_1$. If the mass flow is prescribed and the flow is uniform and attached, continuity gives $U_2 = U_1/R$. Ideal Bernoulli recovery gives $p_2 - p_1 = \frac{1}{2}\rho U_1^2(1 - 1/R^2)$. This is a maximum ideal recovery for that assumed passage

flow, not a whole-car downforce calculation. It cannot tell you how much air enters, where separation begins, or what pressure acts over the floor.

Workbench · Diffuser geometry and ideal recovery

Teaching model: fixed 35 m/s throat flow, constant width, no friction, leakage, or separation. Changing the ramp updates geometry and an ideal recovery only. There is deliberately no predicted car downforce.

Throat height

0.14 m

Ramp length

0.95 m

Ramp angle

7°

256.6 mm

Exit height

1.83

Geometric area ratio

19.09 m/s

Ideal exit speed

70.2%

Ideal recovery fraction

527.0 Pa

Ideal pressure recovery

Not modeled

Predicted car downforce

Ramp 7.0° (schematic)

Floor / throat



Road · geometric expansion only; no solved flow

Angle alone is not a design rule

A longer ramp can attain a given area ratio with a smaller angle, but packaging and friction may change. A steeper ramp can strengthen the adverse pressure gradient and promote separation. Fences can alter crossflow and vortices; their effects depend on floor conditions and can add drag. Widthwise expansion produces another recovery demand. Compare throat height, ramp length, expansion, and upstream state together rather than declaring every 7° diffuser safe or every 14° diffuser stalled.

Near the ground, decreasing ride height can first increase load and later reduce it as the flow loses attachment or supply. The transition can be asymmetric and abrupt. A symmetric half-car steady model can suppress possible left/right states. Experiments on one diffuser-equipped bluff body demonstrate such low-clearance load reduction, but their exact heights and ramp geometry are not transferable as a universal operating map.

The teaching kit uses replacement bodies

The two supplied diffuser shapes are built into alternative body meshes. They remove material that the flat-floor shell otherwise occupies; simply adding a ramp to the baseline shell would leave the original floor blocking the intended passage. Use exactly one body group at a time, while keeping all body part switches on for a shared grid. Inspect the underside before every run. This simple distinction between replacing geometry and adding geometry is essential in real CAD studies.

Lab 06 · 75 min · Flat floor, two ramps, and skirts

Run flat body, 7° diffuser body, 14° diffuser body, and 7° diffuser body with skirts. For each diffuser run switch Flat-floor body OFF and exactly one diffuser body ON. Keep wings, splitter, and canards off. Save centreline and off-centre velocity sections, underside pressure, force history, and total force. Inspect whether the ramp flow and upstream floor suction change together.

Predict: ideal recovery increases with area ratio in the calculator. **Interpret:** explain why the CFD need not follow that ordering. Check mesh resolution, leakage, attachment evidence, and convergence. The supplied results are new synthetic-car examples, not a reproduction of the paper in Lesson 13.

Experimental boundary on the intuition: Senior and Zhang (2001), diffuser-equipped bluff body in ground effect.

DAY 2 / 12 / DEVICES

Ride height, skirts, rake, and cooling

You will identify the conditions that can make a promising floor fragile or incomplete.

Sealing is a pressure-management problem

When underfloor pressure is below the adjacent air, lateral leakage can weaken suction and alter the floor flow. A skirt or fence can impede that leakage, guide flow, or generate a useful vortex. It also adds wetted area and can change wheel interactions. Mechanical skirts close to the road risk contact and must operate over suspension travel and road disturbances. A vortex-based seal depends on the flow maintaining its position and strength. Neither is immune to yaw or changes in clearance.

The supplied paper investigates diagonal sealing skirts as part of its particular floor. Those are different from a generic vertical side skirt or a side sill extension. The course pack uses two simple side fences as a teaching variation. Their results cannot claim to reproduce the paper's diagonal-skirt performance. Read the geometry specification before comparing a device name across studies.

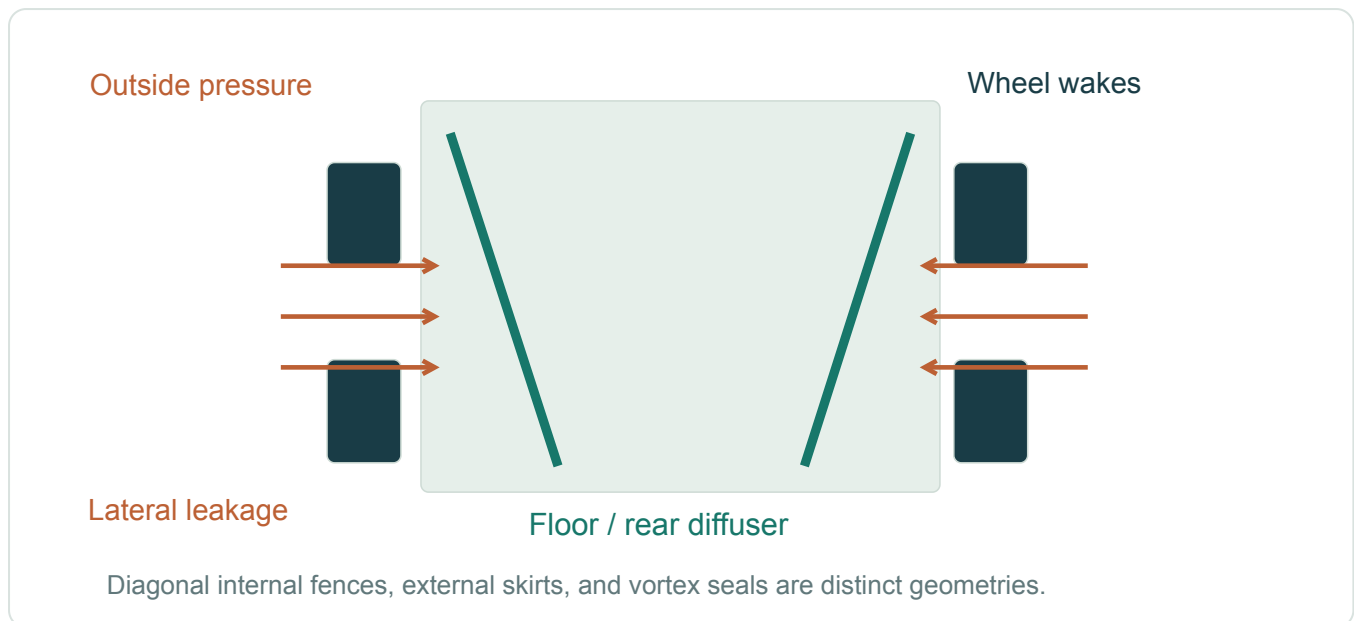


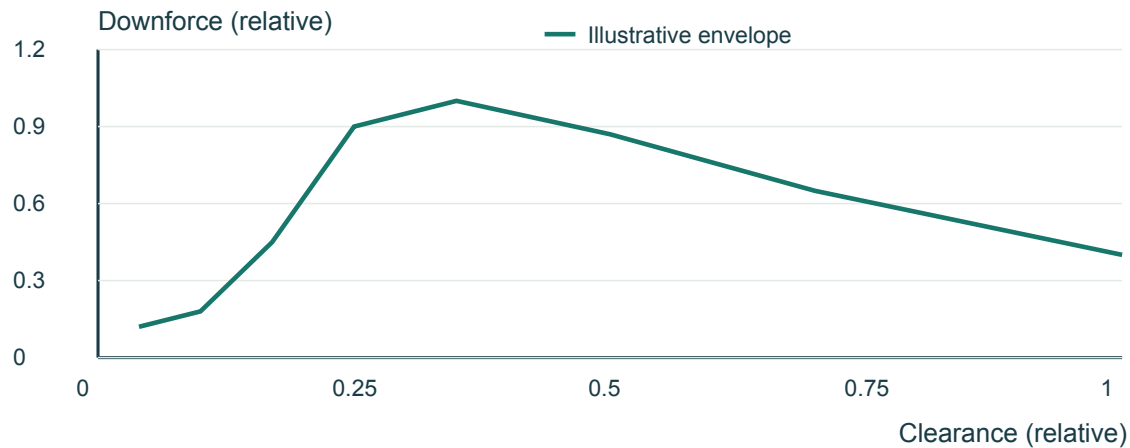
Figure 15. Plan-view leakage paths, wheel wakes, and possible sealing concepts. Different skirt geometries are distinct experiments.

Ride height and rake form an operating map

Front and rear ride heights define a floor's pose. Rake is the longitudinal inclination; for small angles it is approximately $(h_{\text{rear}} - h_{\text{front}})/\text{wheelbase}$. Changing rake changes the entry gap, underfloor expansion, diffuser relative angle, body incidence, and wing incidence together. It is not merely a diffuser-angle adjustment. An aero map records forces and moments across these conditions so the suspension can operate within a stable range.

A useful map might start with front/rear heights of 80, 110, and 140 mm for an imagined car, but those are study parameters that must fit its geometry and contact constraints. Do not translate the entire car down until a tyre penetrates the road. In the browser app a road-height adjustment represents a global geometric shift, not a suspension model with wheel centres fixed and the sprung body pitching. Correct pose studies require carefully prepared geometry variants or a dedicated pose tool.

Possible low-clearance load loss / schematic



A low-gap peak can be followed by flow separation or reduced supply.

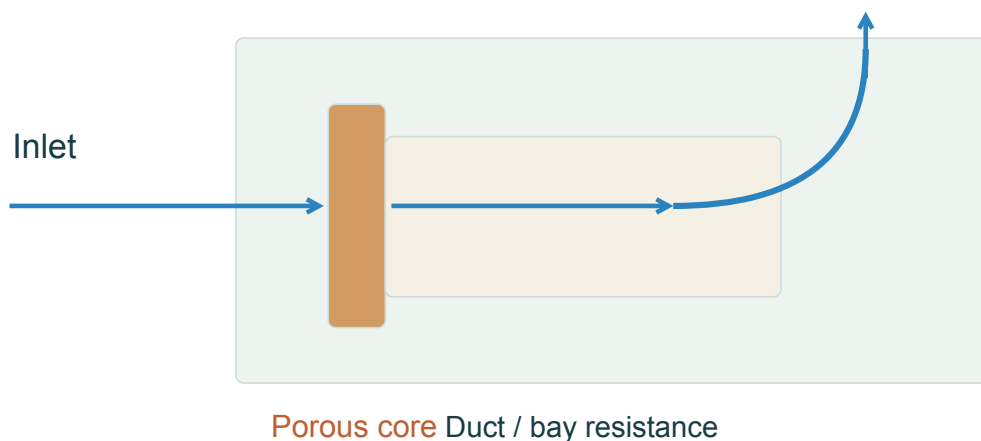
Figure 16. An illustrative non-monotonic downforce-versus-clearance curve. It shows a possible mechanism, not measured values or universal thresholds.

Cooling flow belongs in the drag and lift budget

Air passing through a radiator loses total pressure. Inlets and outlets also change the outer body pressure field. Closing a grille can reduce drag in a geometry model but fail the thermal duty. A hood outlet may reduce engine-bay pressure and front lift when correctly placed; it can also disrupt windshield flow or worsen drag. You need mass flow, pressure drop, heat rejection, and the external flow interaction to make a cooling decision.

For a first-order porous model, Δp may be represented as $K_1 U + K_2 U^2$ with coefficients fitted to a core's measured pressure drop. Couple that with appropriate inlet and outlet passages and a thermal requirement. An external-only sealed-body simulation cannot assess radiator airflow. The present course therefore treats cooling as a missing physical subsystem when the geometry contains no ducts or core. The cooling-flow feature request specifies the additional outputs needed to study this tradeoff.

Mass flow and pressure loss must be modeled



Closing an external opening is not a cooling-capability result.

Figure 17. A cooling flow path needs an inlet, a resistance model, and an outlet. External shape alone does not specify its mass flow.

End-of-day package audit

By now each device should have a mechanism, a likely cost, and an operating sensitivity in your notebook. For a wing: local incidence, profile and induced drag, and structural deflection. For a splitter: pressure difference, contact risk, and front balance. For canards: direct load, downstream interaction, and yaw sensitivity. For a floor/diffuser: leakage, pressure recovery, and clearance dependence. For skirts: sealing, drag, and a physically viable gap. If your notes only say “more downforce,” add the mechanism and the test that could disprove it.

▼ Day 2 checkpoint: why is a sealed grille or a 5 mm skirt gap not automatically a valid improvement?

A sealed grille changes thermal capability, while a tiny skirt gap may disappear under suspension motion or be unresolved numerically. A useful package must meet cooling, clearance, structural, and aerodynamic requirements together.

Research case study: the diffuser paper

You will read a published CFD study critically, recompute its comparisons, and distinguish a research finding from a transferable design rule.

The paper supplied for this course is Guerrero, Castilla, and Eid (2022), A Numerical Aerodynamic Analysis on the Effect of Rear Underbody Diffusers on Road Cars, Applied Sciences 12, 3763, DOI 10.3390/app12083763. It is open access under CC BY 4.0. The table below transcribes Table 3; the diagram is an original schematic, not a copy of the CAD or a new simulation of the paper.

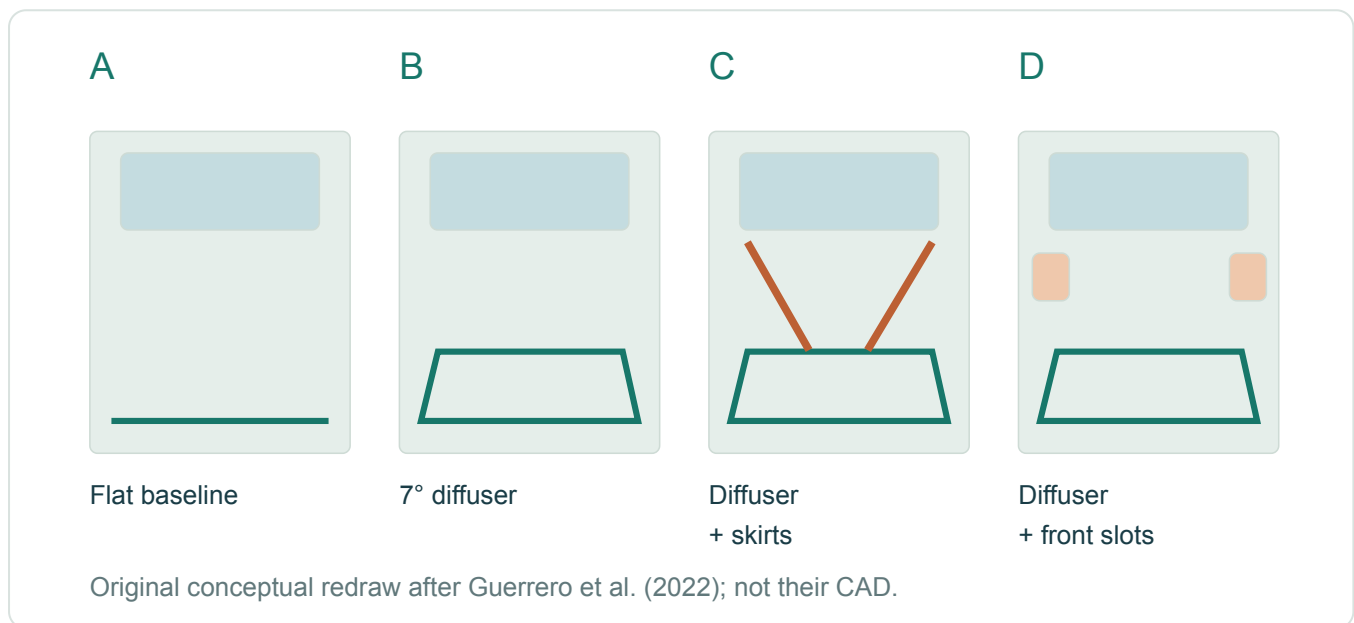


Figure 18. Conceptual differences between the four published configurations. Original diagram after Guerrero et al. (2022), CC BY 4.0; not dimensionally faithful CAD.

What was actually studied?

The authors used a self-designed road-car geometry about 4 m long with approximately 2.45 m wheelbase. The configurations were a flat-underfloor baseline, a rear 7° diffuser, that diffuser with diagonal sealing skirts, and a diffuser configuration with front slots near the tyres. The front-slot case is a separate branch; it should not be read as "skirts plus front slots." Their numerical method was steady incompressible RANS in OpenFOAM with simpleFoam and $k-\omega$ SST. The study used a tetrahedral mesh with local refinements and boundary layers; it is not the same mesh pipeline as EasyCFD's Cartesian browser solver or its snappyHexMesh backend.

The inlet and moving ground were set to 25 m/s, equivalent to 90 km/h. Wheels used rotating-wall conditions at 87 rad/s. The density was 1.225 kg/m³, turbulence intensity 0.150%, length scale 0.172 m, and reported Reynolds number 4 million. Only half the car was simulated with symmetry; remaining side/top boundaries used slip. The finest mesh had 6.7 million elements. Convergence was judged using residuals below 10^{-5} and stabilised coefficients at around 1000 iterations. Each of these conditions limits what can be inferred about another setup.

The mesh description reports nine wall layers with a growth factor of 1.5, a first-cell height of 0.01 mm, and an averaged y^+ of 40. These are reported quantities, not values measured by this course. Audit the units and wall-distance convention before copying them. For example, with an assumed friction velocity of 1 m/s and $\nu = 1.5 \times 10^{-5} \text{ m}^2/\text{s}$, the centre of a 0.01 mm-high first cell would give $y^+ \approx 0.33$. That conditional arithmetic does not reproduce the authors' solution; it shows why actual local distances and y^+ distributions are needed to reconcile the reporting.

Published results · Guerrero et al., Table 3

Configuration	Cd	Cl	Positive downforce / drag, -Cl/Cd	Δ Cd vs baseline	Δ positive load coefficient
Flat baseline	0.320	-0.047	0.147	0	0
7° diffuser	0.325	-0.573	1.763	+0.005	+0.526
Diffuser + diagonal skirts	0.324	-0.887	2.738	+0.004	+0.840
Diffuser + front slots	0.332	-0.773	2.328	+0.012	+0.726

The efficiency column here is calculated directly from the displayed coefficients with positive downforce. The paper reports signed $E = Cl/Cd$: -0.146, -1.763, -2.737, and -2.323. Small differences from the recalculated values may reflect rounding or reporting; keep the signed published values separate from a newly computed ratio. This is an example of a numerical audit, not grounds to silently change the paper's source data.

Turn the headline into transparent arithmetic

The skirt configuration changes the downforce coefficient magnitude from 0.047 to 0.887. The increase is $(0.887 - 0.047)/0.047 \times 100\% \approx 1787\%$, consistent with the paper's "around 1780%." The final load is about 18.9 times a small baseline. The huge percentage is partly a consequence of that small denominator. The corresponding Cd increase is $0.004/0.320 = 1.25\%$. Both the absolute differences and the denominator are needed to understand the tradeoff.

At the paper's speed, $q = 382.8125 \text{ Pa}$. Without assuming the study's reporting area, dimensional changes per square metre of that area are $\Delta D/A_{\text{ref}} = 1.531 \text{ N/m}^2$ and $\Delta N/A_{\text{ref}} = 321.56 \text{ N/m}^2$ for the skirts case. To convert to total newtons you need the exact area and coefficient definitions used by the authors. Do not substitute your own car's frontal area into a published coefficient and call the result a prediction for your car.

What evidence does the paper provide, and what remains open?

The paper reports a three-mesh GCI analysis on baseline drag, with reported fine GCI 1.218%, coarse GCI 2.928%, and an asymptotic-range parameter 1.032. This supports the authors' numerical refinement argument for that metric and setup. It does not separately quantify lift, moment, each modified configuration, modeling error, or real-car accuracy. The rounded displayed coefficient sequence cannot reproduce every reported GCI quantity without further numerical detail; request unrounded values and explicit grid ordering before treating it as a worked arithmetic reference.

The authors explicitly propose experimental validation and unsteady simulations as future work. Their strong downforce gains are therefore numerical findings for their modeled geometry and conditions. Symmetry precludes a freely developing left/right asymmetric state. The near-wall values and layer dimensions should be checked in an attempted reproduction using actual local y^+ distributions, not inferred from a single averaged number. A smoother-looking wake and modest C_d increase do not prove that every diffuser has cleaner flow or lower drag than every wing.

Lab 07 · 60 min · Audit and transfer a research result

Recompute ΔC_d , ΔC_l , efficiency, and the 1780% headline. Identify five mismatches between the study and your teaching car: geometry, mesh method, speed/Re, turbulence input, and symmetry are useful starting points. Design a new study inspired by the paper, holding your own conditions fixed. State what would count as a reproduction and what is only a related experiment.

Deliverable: a source-data table, your calculations, and a transfer statement: "This suggests testing coherent floor flow and sealing on my geometry; it does not establish that 7° or diagonal skirts will give the same load." Explain why proving a small drag penalty can be harder than detecting a large lift difference.

DAY 3 / 14 / METHODS

What the CFD solver actually solves

You will understand the modeling hierarchy sufficiently to choose an appropriate question and interpret solver output.

Mass and momentum are the foundation

For an incompressible Newtonian fluid, continuity requires $\nabla \cdot \mathbf{U} = 0$. Momentum balances acceleration with pressure forces and viscous stresses. At high Reynolds number, a direct simulation of all turbulent scales around a full car is extremely expensive. Engineering CFD normally models at least part of turbulence. The important distinction is what the chosen method resolves, what it averages, and what its additional closures assume.

$$\nabla \cdot \mathbf{U} = 0$$

$$\rho[\partial \mathbf{U} / \partial t + (\mathbf{U} \cdot \nabla) \mathbf{U}] = -\nabla p + \nabla \cdot \boldsymbol{\tau}$$

RANS adds modeled stresses from unresolved velocity fluctuations.

Reynolds-averaged Navier-Stokes uses a mean velocity and additional stresses from turbulent fluctuations. An eddy-viscosity model approximates those stresses using the local mean strain and a turbulence viscosity. k - ω SST transports turbulent kinetic energy k and a specific dissipation variable ω , combining near-wall and

outer-flow behaviour through blending functions. It is a useful engineering closure, not a guarantee of correct separated flow for every car.

Numerical convergence and physical accuracy are different

A discretisation replaces the continuous equations by algebraic relations on a mesh. Pressure-velocity coupling enforces mass conservation while momentum is advanced or iterated. Linear-solver residuals, nonlinear residuals, and force settling are related but distinct diagnostics. You can solve a poor mesh or inappropriate model very consistently. Refining the mesh reduces a class of numerical errors but does not remove a missing cooling passage, an incorrect road boundary, or a turbulence-model bias.

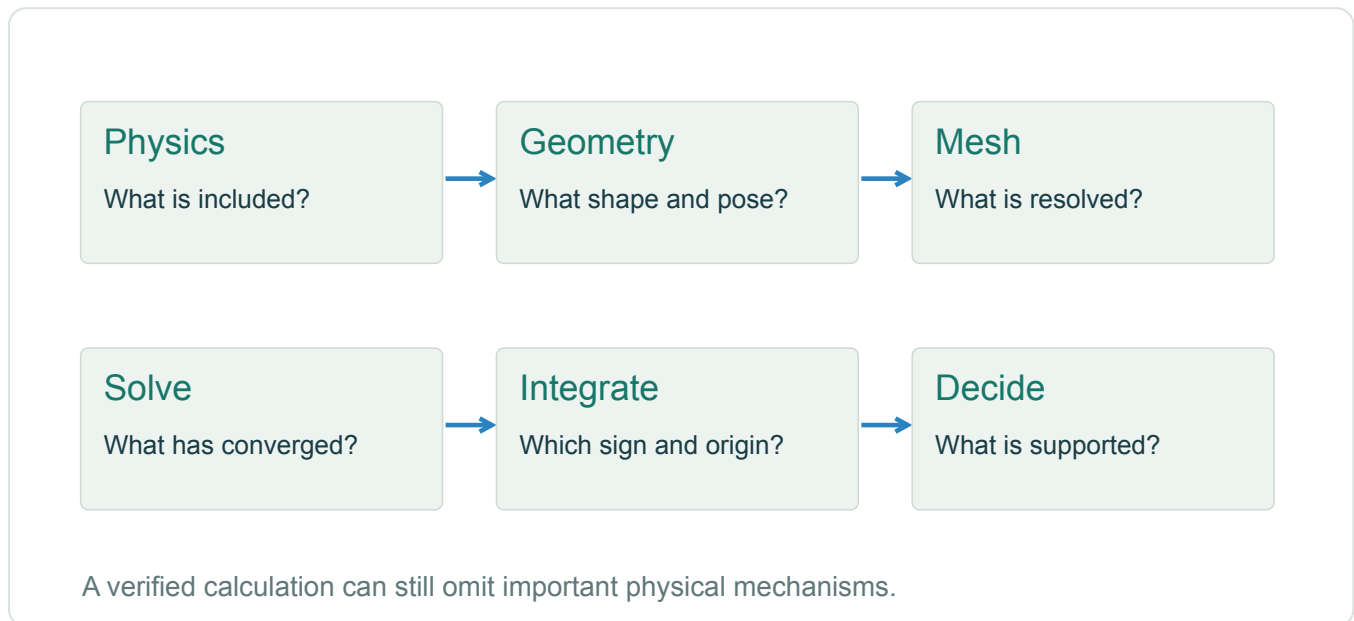


Figure 19. A prediction passes through physical assumptions, geometry, discretisation, iterative solution, integration, and interpretation. Each stage can introduce error.

Steady RANS seeks a stationary mean solution. Local time stepping can accelerate convergence by giving different cells different effective step sizes. That progression is pseudo-time, not a physical time record. URANS uses time-dependent averaged equations; LES resolves a larger share of unsteady structures while modeling smaller scales. These approaches require appropriate temporal and spatial resolution and sampling. Switching labels to “transient” does not establish a validated unsteady-load prediction.

Two EasyCFD engines, two numerical treatments

The browser solver uses a stretched Cartesian cut-cell formulation, thin-part handling, SST turbulence, and wall modeling. The OpenFOAM path uses the existing backend's snappyHexMesh/prism-layer/simpleFoam pipeline. Their discrete operators, mesh geometry, wall representation, sampling, and convergence can differ even when the broad physical model and inputs sound alike. Agreement between two unqualified solvers is reassuring evidence about a comparison, but it is not experimental validation. Disagreement does not tell you which is physically right without further evidence.

The repository documents significant uncertainty for lift and small aero devices, including cases where browser wing downforce is under-predicted or the incidence ordering differs from OpenFOAM. Fast, Medium, and Precise are effort presets rather than accuracy certifications. The two-grid averaging in the browser Precise preset is not a substitute for three independent converged meshes and an observed-order study. Use Custom cases and export histories for a deliberate refinement investigation.

State a prediction's scope before running it

An external incompressible rigid-car calculation at one speed and zero yaw can answer a narrow question about that modeled condition. It cannot establish thermal adequacy, body motion over bumps, tyre deformation, structural flexibility, acoustics, or gust response unless those mechanisms are included and qualified. A responsible scope statement is practical: it prevents you from spending time resolving detail that cannot answer your actual design question.

▼ Check: your residuals are low and the fields look smooth. Have you validated the diffuser?

No. You have evidence about iterative behaviour. You still need force/moment settling, mesh and domain sensitivity, appropriate physical assumptions, and comparison with suitable experimental evidence.

Implementation details and numerical evidence: EasyCFD validation report. Turbulence context: NASA Turbulence Modeling Resource.

DAY 3 / 15 / METHODS

Geometry, boundaries, and the virtual tunnel

You will prepare a controlled external-flow case and identify boundary-condition mistakes before solving.

Geometry is an engineering input

Check dimensions immediately after importing. STL does not encode units, so a millimetre model interpreted as metres can invalidate every setting. Confirm the nose and up axes, body/wheel identification, wheel radius, clearance, missing faces, duplicate shells, and the enabled groups. Keep variants in one common coordinate frame. Export a stable naming scheme and record which surfaces were removed or replaced. A diffuser ramp inside an unchanged closed body is not the intended geometry.

Closed, outward-oriented volumes are generally easier to interpret than fragmentary surfaces. Thin parts may be handled as zero-thickness walls by the browser solver, which is a numerical representation rather than a full resolution of thickness and edge structure. Very small gaps, slotted elements, and internal channels need explicit fidelity checks. Geometry-rendering quality and mesh quality are separate. Merging or sealing a mesh can alter slots and tyre gaps; review the resulting geometry before running.

Watertightness and orientation are separate checks. Two triangles can meet along an edge while both traverse it in the same direction, leaving one face reversed. Inspect consistent adjacency and outward orientation per closed component, not only the absence of holes. The course generator verifies finite vertices, two opposite directed incidences per edge, and positive signed volume, and records file hashes. The app's current undirected open-edge count does not provide all of those orientation checks.

Use the road frame deliberately

At zero yaw in a car-fixed frame the road moves rearward with the road-relative speed, and the tyre angular velocity should match rolling approximately through $\omega = U_{\text{road}}/R$. The body's no-slip velocity is zero. A stationary floor creates a different floor boundary layer from a moving road. Non-rotating wheels also change the flow. These can be useful controlled sensitivity experiments, but they do not represent a car driving over a road.

Crosswind complicates this frame: road motion and wheel rotation are set by vehicle motion over the road, while inlet direction and magnitude are set by air motion relative to the car. A lateral wind does not make the road slide sideways. Audit the solver's implementation when interpreting yaw cases; use the wind triangle in Lesson 22 to separate road speed and airspeed. An app with only one speed field cannot automatically express every ambient-wind condition.

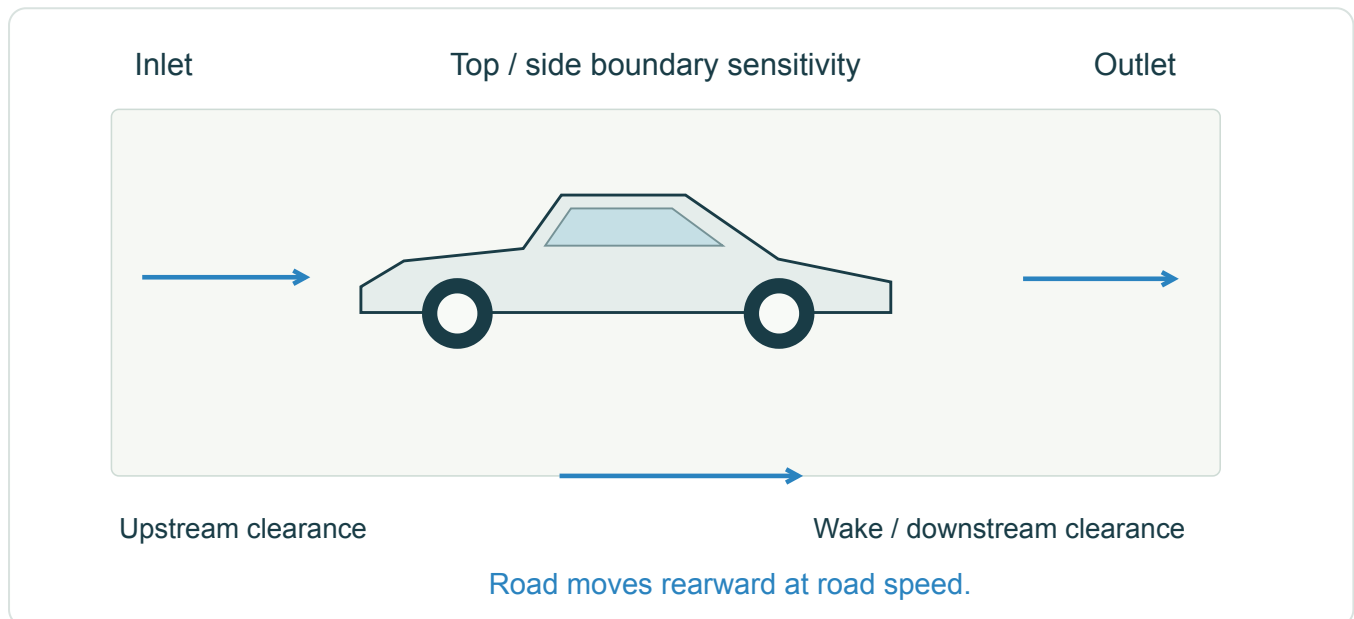


Figure 20. Domain boundaries and car-fixed road motion at zero yaw. The extent ratios are study starting points, not an accuracy guarantee.

Keep artificial boundaries far enough away

A velocity inlet prescribes approaching flow; a pressure outlet allows the downstream pressure level to anchor the solution. Side/top boundaries and the ground influence the whole pressure field if too close. Blockage is projected frontal area divided by the tunnel cross-section. A small blockage ratio helps, but a single threshold cannot certify negligible confinement. The rear wake needs sufficient length for outflow conditions to avoid distorting the region of interest.

As a starting experiment, consider several car lengths upstream, more downstream, and generous lateral/top clearance; the necessary values depend on boundaries and the question. Expand the domain and compare forces and moments while maintaining comparable near-car resolution. If the domain changes the result more than the modification does, the design ranking is not resolved. Record actual domain bounds rather than only a vague phrase like "large tunnel."

Fixed conditions make variants comparable

Use an explicitly declared density, freestream turbulence model/input, reference area, road and wheel treatment, orientation, and pose. Keep the complete variant family present in the shared-grid design. A group switch turns a geometry variant on for the solve while allowing inactive variants to keep shaping the common grid. Turning the part's own switch off removes it from that grid definition. Check the saved grid identifier instead of assuming two runs used identical cells.

Lab 08 · 60 min · Boundary and domain sensitivity

Duplicate a baseline design. Test a deliberately enlarged simulation box with the same near-car custom resolution, and document the actual resulting grid and cell counts. Separately compare stationary versus moving road with all other settings held fixed. These are two independent studies; do not change road and domain together. Record Cd, signed Cl, moment if available, histories, and underside flow.

Deliverable: identify which difference is a physical modeling choice and which estimates numerical boundary sensitivity. Neither exercise demonstrates that a different boundary is a better design.

DAY 3 / 16 / METHODS

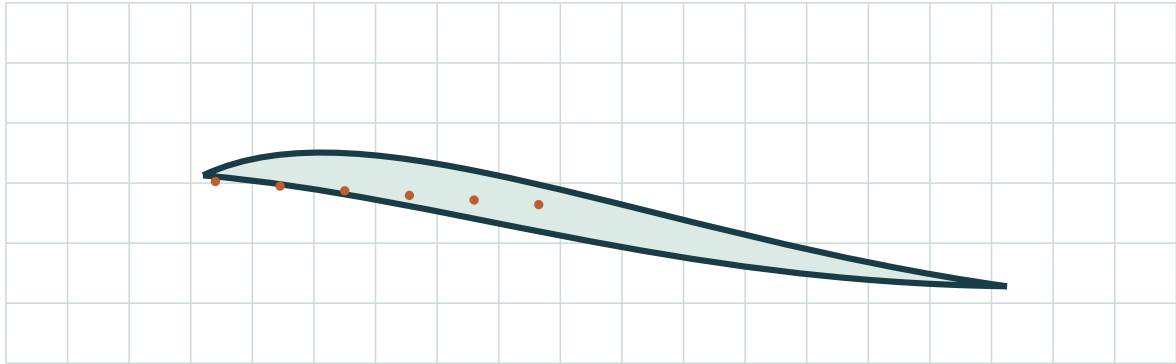
Meshes, y^+ , convergence, and uncertainty

You will build a verification argument and avoid reporting a precise-looking number with unjustified precision.

Refine where the relevant gradients live

Leading edges, thin wings, floor gaps, diffuser ramps, wheel contact regions, and separated shear layers need appropriate resolution. A global cell count alone does not describe that resolution. Report chordwise/spanwise sampling, near-wall layers, smallest effective gap representation, and wake refinement. On a cut-cell grid, small shifts of the geometry relative to cells can alter integration and geometry representation. Paired studies on a shared grid reduce one comparison difference but do not prove the design delta is mesh independent.

Smooth geometry is not the same as resolved flow.



Small leading-edge radii, gaps, and boundary layers need their own resolution.

Figure 21. A smooth CAD wing can still have inadequate numerical resolution. Geometry triangles, solver cells, and visualisation samples are different objects.

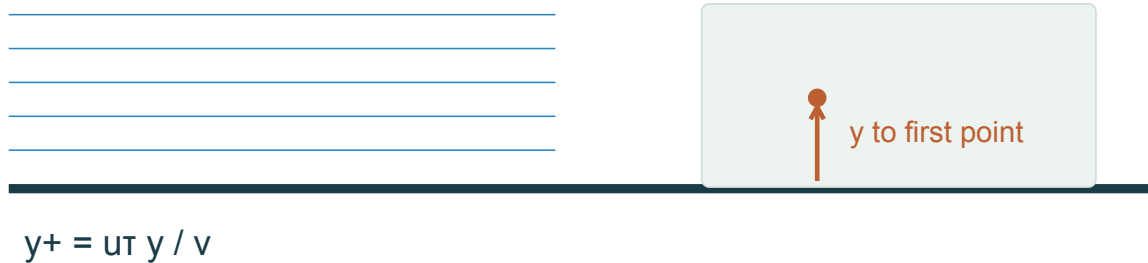
Near-wall y^+ depends on the solution

Define $y^+ = u\tau y/\nu$ with friction velocity $u\tau = \sqrt{(|\tau_{\text{wall}}|/\rho)}$ and y the representative wall-normal distance of the first cell point. A wall-resolved treatment and a wall-function treatment have different requirements. Order-one y^+ is a typical goal for a genuinely wall-resolved approach; log-layer wall functions generally need their appropriate log-layer range and adequate layer coverage. Blended wall models are not permission to ignore near-wall resolution. Values vary over the car and require inspection, not just a target entered before solving.

Use the distance convention from the solver. If a prism's full height is used instead of its centre-to-wall distance, an estimate can differ by about a factor of two. Check actual y^+ distributions on the wing, floor, and ramp, not only the global average. Near separation τ_{wall} tends toward zero, so a small y^+ there alone does not demonstrate excellent boundary-layer resolution. Layer collapse and insufficient layers can remain serious problems.

Wall-resolved inner layer

Modeled wall relation



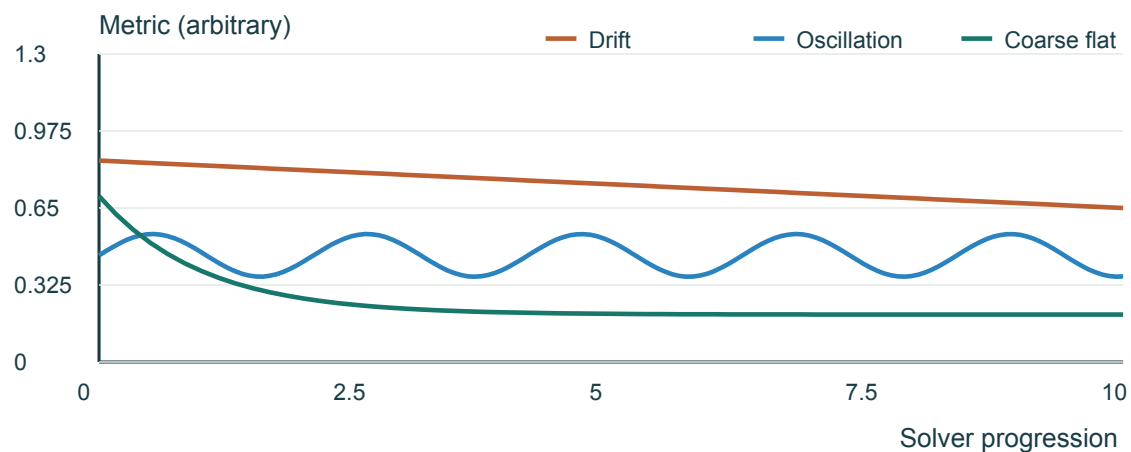
Inspect local y^+ and actual layer coverage, not only an average.

Figure 22. Representative wall distance and the distinction between resolving and modeling the inner layer. The drawing is conceptual.

Three levels of convergence

First, iterative convergence: algebraic and nonlinear residuals and mass conservation must be suitable for the case. Second, force/moment convergence: means and trends must be stable over a justified window, or a physically appropriate unsteady average must be sampled. Third, discretisation convergence: repeat on finer, comparable grids. Also test domain sensitivity. Each test addresses a different uncertainty source. Their absence cannot be compensated by an extra decimal place.

Three traces / different interpretation



Flat force history alone does not demonstrate mesh or physical accuracy.

Figure 23. Illustrative force-history behaviours: drift, stationary oscillation, and apparently flat coarse output. These synthetic traces are not course CFD runs.

A narrow but useful three-grid calculation

For geometrically similar grids with a common spacing ratio $r > 1$, label ϕ_1 fine, ϕ_2 medium, ϕ_3 coarse. For a monotonic asymptotic sequence, $p = \ln|(\phi_3 - \phi_2)/(\phi_2 - \phi_1)| / \ln r$. Richardson extrapolation gives $\phi_{\text{ext}} = \phi_1 + (\phi_1 - \phi_2)/(r^p - 1)$. A common fine-grid GCI is $1.25 |(\phi_1 - \phi_2)/\phi_1|/(r^p - 1)$. This factor assumes a three-grid analysis and is a reporting convention, not a universal statistical confidence level.

Workbench · Three-grid verification arithmetic

Teaching calculation for constant spacing ratio, monotonic converging data, and a nonzero metric. Invalid sequences are rejected. The result estimates discretisation uncertainty under these assumptions, not model or experimental uncertainty.

Fine-grid metric	0.3	Medium-grid metric	0.306
Coarse-grid metric	0.3195	Constant spacing ratio r	1.5

2.000

Observed order p

0.29520

Extrapolated metric

2.00%

Fine-grid GCI

Requires comparable, converged grids in an asymptotic regime. A 2.00% GCI is not a physical-accuracy claim.

If differences change sign, grow with refinement, vanish through rounding, or involve incompatible meshes, do not force the formula. Lift near zero requires an absolute uncertainty scale rather than a misleading relative percentage. Local/adaptive refinement and a two-grid averaged preset do not automatically provide the necessary constant- r sequence. Analyse drag, lift, moment, and design differences independently. A grid study on baseline C_d does not certify the wing's C_l .

How small a design difference can you trust?

Suppose $\Delta C_d = -0.004$ but plausible variations from force sampling and grid phase are of order 0.010. You have not resolved that gain. Paired differences can share correlated error, but assuming cancellation without evidence is optimistic. Repeat the comparison across mesh levels and reasonable shifts, quantify the delta itself, and use a physically validated reference when needed. The current app's temporal bands describe window behaviour only; they do not include geometry, mesh, boundary, turbulence-model, or measurement uncertainty.

Method background: NASA/NPARC spatial-convergence tutorial. Near-wall output: OpenFOAM yPlus documentation.

DAY 3 / 17 / PRACTICE

Run controlled device experiments in EasyCFD

You will create a repeatable experiment with documented variants and a result that can be reviewed later.

The independent teaching pack

The course includes an original synthetic 4.2 m teaching car with four cylindrical wheels, flat-floor and two integrated-diffuser body versions, two inverted NACA 4412 wing versions, a front splitter, flat teaching canards, and side skirts. All parts share one coordinate frame. The body throat is 140 mm above the road, the ramp is 950 mm long, wing chord is 320 mm, span is 1.7 m, and wheel radius is 320 mm. Wing supports, realistic wheelhouses, detailed cooling, and manufacturing interfaces are omitted. This is mechanism geometry, not a fitment model.

The app link creates a new saved course design with optional groups initially off. Existing designs and runs remain in the library. The offline STL pack includes import instructions. For manual import, import the flat body and four wheels first, then use Add parts for the rest. Rename the automatic groups to the names shown here. The browser app does not need a backend for these WebGPU exercises.

Exactly one body; at most one wing.

The diffuser meshes replace the flat body. For a diffuser case switch the Flat-floor body group off and one diffuser body group on. Do not use "Only this" without checking the body: that action intentionally keeps the base body active. Use group switches for variants, and retain each part's own switch so the common grid remains unchanged. Course meshes are not a valid overlapping-shell combination.

The experiment matrix

Run names and enabled geometry

Run	Body	Other groups	Question
B0	Flat	Wheels only	What is the baseline flow and uncertainty?
W6 / W12	Flat	One wing version	How does installed incidence affect forces?
S	Flat	Splitter	Does front pressure difference create useful load?

Run	Body	Other groups	Question
C	Flat	Canard pair	How much is direct versus downstream change?
D7 / D14	One diffuser body	Wheels only	Does more geometric recovery improve actual flow?
D7K	7° diffuser	Side skirts	How does leakage control affect the floor?
P	7° diffuser	Wing 6°, splitter, canards, skirts	Are separate device gains additive?

Set 100 km/h, zero yaw, density 1.225 kg/m³, Aref = 2.2 m², moving ground and rotating wheels. For comparison with the included runs choose WebGPU Custom, 72 cells per car length, 10 passes, detail ratio 1/off. Use the same domain choice and keep all variant parts available. Save the actual cells, grid identifier, solver version, run warnings, and any automatically extended passes. “72 cells” does not mean only 72 total cells; transverse, vertical, and expanded-domain cells can make the total much larger.

Use the interface as an experimental notebook

In Car, verify dimensions, axes, roles, group switches, and road gap, then confirm the checklist. In Conditions, record speed, yaw, density, area, road/wheels, and domain. In Run, record engine, effort, detail cells, and animation choices. In Results, inspect the history and warnings before the mean coefficients. Explore airflow using pressure, surface flow, horizontal/vertical streamlines, and wake views. In Compare, use shared scales and synchronized cameras, then export JSON, CSV, and selected PNG views.

After a variant, compare the whole-car force with the sum of available group contributions and inspect reconciliation warnings. OpenFOAM currently has different group-breakdown capabilities; do not invent missing per-group forces. Keep a description of the geometry differences with the run. A file name alone is insufficient if its contents changed between runs.

Lab 09 · 90 min · Verify a design difference

Pick one pair from the matrix, such as B0 and S. Repeat both at custom resolutions of 60, 72, and 90 cells per length using a sufficiently long common averaging plan. These are separate single-resolution runs, not the Precise average. Extend unsettled cases if feasible and record failures. Compare ΔC_d , ΔC_l , and pitch/axle-load changes across levels; do not fit a GCI if the actual grid refinement is not compatible.

Deliverable: a six-run comparison, an uncertainty discussion, and one of three conclusions: the difference is resolved within this model; the difference is unresolved; or the study failed to provide usable output. All three can be valuable results.

DAY 3 / 18 / EVIDENCE

Read the real course runs and diagnose a result

You will evaluate actual solver output without turning a numerical demonstration into an engineering recommendation.

The following table and history explorer use raw output captured from the browser solver at the recorded base commit on the original course meshes. They are genuine WebGPU computations. They are not the paper's car, not a calibrated experiment, and not a qualified design ranking. Their source JSON records the capture time, base commit, device adapter, settings, geometry recipe, histories, grid identifiers, cell counts, temporal bands, and warnings. Download it to audit the examples.

Captured 2026-10-07 · source commit b5816e2f831e · apple metal-3 · 72 cells/length, 10 requested passes · 100 km/h · Aref 2.2 m². Source: <web/course/evidence/runs.json>.

Actual exploratory course-run output · signed coefficients, saved means

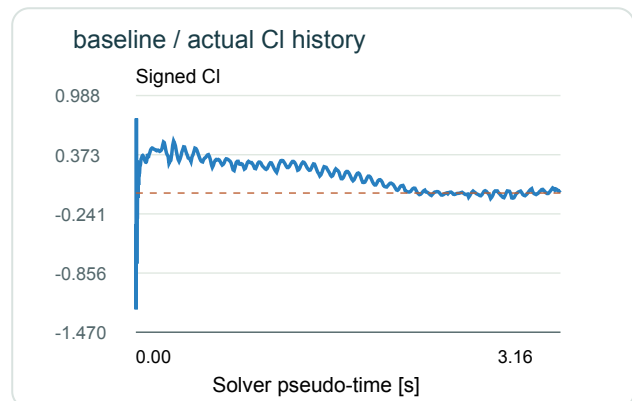
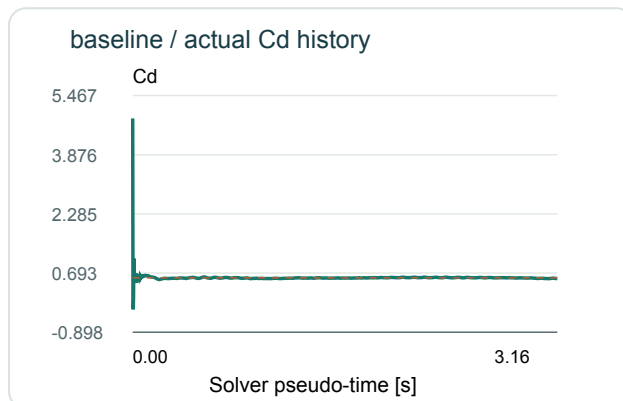
Case	Cd	Cl	Δ downforce [N]	Cells	Settled?	Runtime [s]
baseline	0.5650 ±0.0122	-0.0246 ±0.0196	0.0	1,474,560	No	198
wing6	0.5475 ±0.0012	0.0805 ±0.0049	-109.2	1,474,560	Yes	171
wing12	0.5486 ±0.0016	0.0786 ±0.0096	-107.3	1,474,560	Yes	150
splitter	0.5756 ±0.0073	-0.1644 ±0.0180	145.4	1,474,560	No	188
canards	0.5386 ±0.0145	0.0552 ±0.0271	-83.0	1,474,560	No	197
diffuser7	0.5876 ±0.0148	-0.0926 ±0.0963	70.7	1,474,560	No	223
diffuser14	0.5718 ±0.0112	0.0121 ±0.0518	-38.1	1,474,560	No	346
diffuser7skirts	0.6213 ±0.0290	-0.3266 ±0.0778	314.0	1,474,560	No	349
package	0.6534 ±0.0535	-0.4363 ±0.0757	428.1	1,474,560	No	321

Bands are half-ranges of sub-window means. “Settled” describes temporal behaviour only. Wall runtimes reflect the capture machine and its GPU load; they are not a controlled performance benchmark. Whole-car wing results here do not establish a valid real-wing ranking.

Workbench · Real EasyCFD run histories

Actual captured CFD data. Histories use solver pseudo-time and final-window averages; the bands are not confidence intervals. No fitted or invented CFD values are shown.

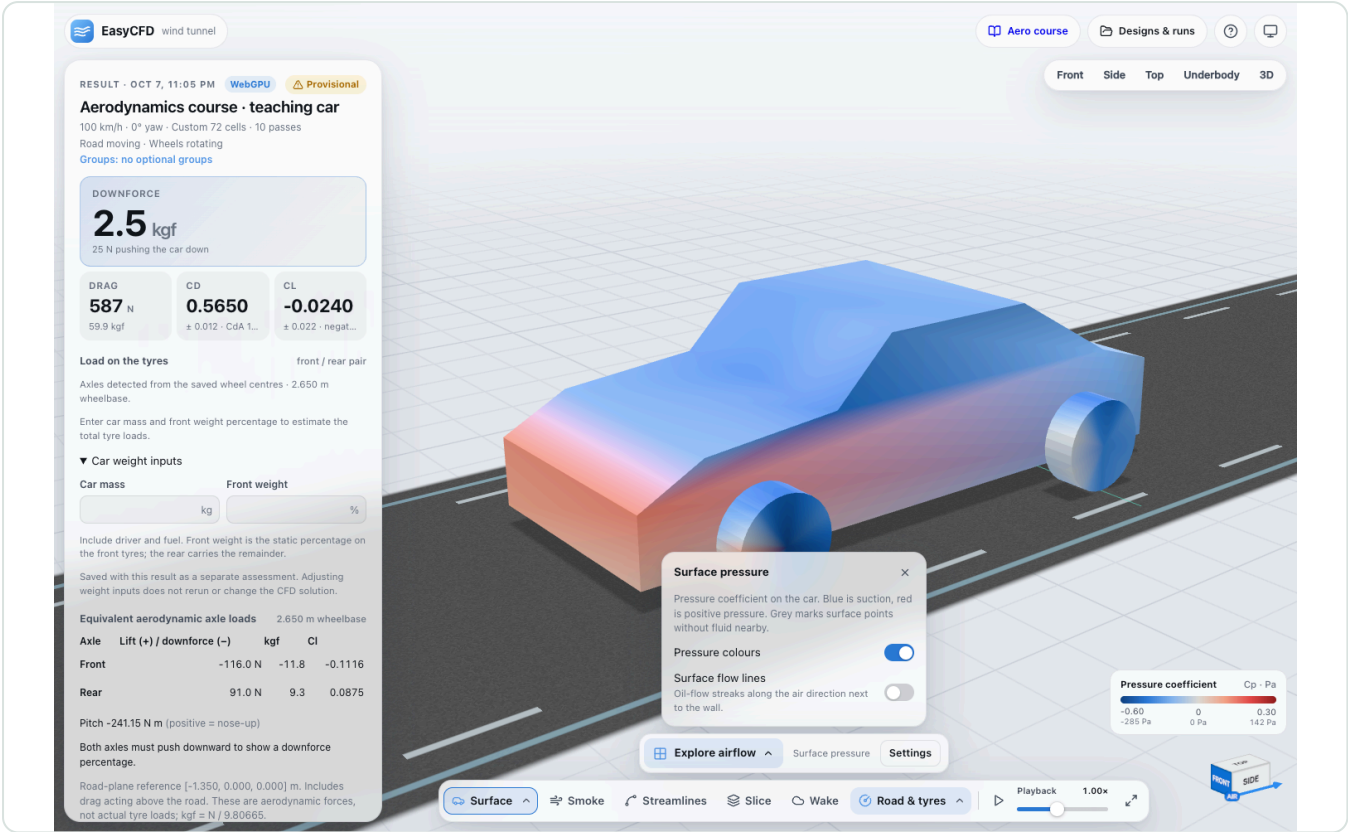
Captured run baseline ▼



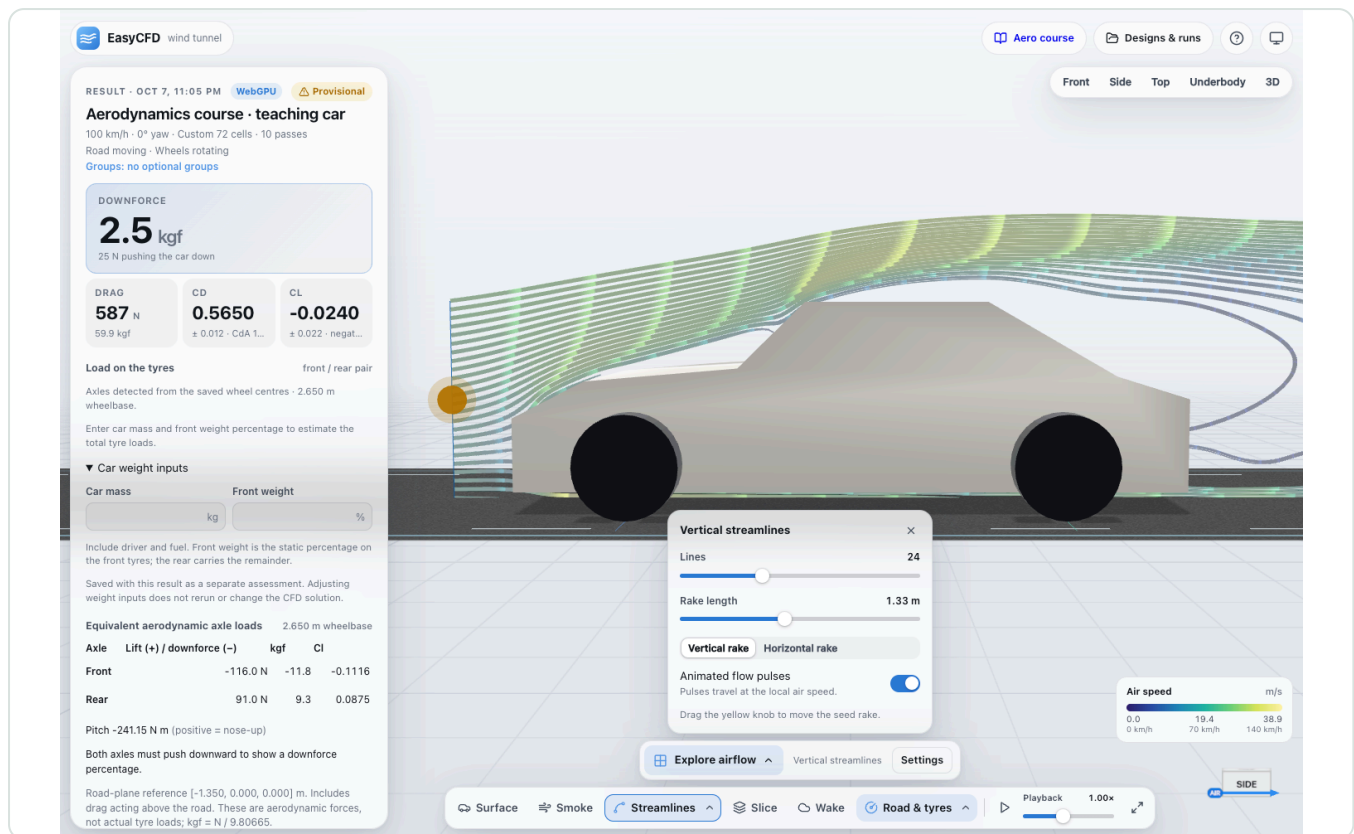
Temporal settling test did not pass. The final-window mean is provisional. Dashed line: saved averaged coefficient. Forces were still drifting, so the run was extended by 10.0 flow passes. They were still drifting at the maximum length; treat them as provisional.

Read two actual field views

These images come from a separate baseline run through the app's own lab launcher and Results view. It used the same 100 km/h, zero yaw, 72 cells/length, 10-pass settings and 1,474,560 cells. Its final-window Cd = 0.5650 and CI = -0.0240 differ from the table's baseline: a second numerical realization is not an exact repetition of every sampled value. This run also failed temporal settling. Its final field illustrates what the app displays; it is not a time-averaged pressure map or experimental measurement. The saved run document, including settings, source hashes, histories, and forces, is included as `viewer-run.json` in the served course resources. Volume-field buffers are not embedded in that JSON; use the capture recipe to regenerate the views.



Field view A. Actual app screenshot, 7 October 2026. Read the pressure-coefficient legend before interpreting colours: red is positive pressure and blue is suction. Relate the nose's high pressure to drag, then remember that suction on upper surfaces can contribute upward lift. Faint optional parts are inactive geometry ghosts; they do not contribute to this baseline solve.



Field view B. The same captured baseline field shown with a vertical streamline rake. The colours now encode local speed, not pressure. Look for the accelerated upper passage and rear recirculation. Streamlines are integrated through a stored field; the animated pulses used by the viewer do not make this an unsteady physical measurement. Move the rake and inspect other planes before inferring a three-dimensional wake.

A repeatable diagnosis sequence

First check that the intended groups, pose, speed, area, and solver were used. Second check initialization errors, NaNs, mesh warnings, blockage, and device limits. Third read all force histories, including lift and moment where available. Fourth inspect the flow hypothesis with surface and volume views. Fifth repeat the key difference with altered resolution or domain, then compare to physical data or a qualified reference. This order saves time because it catches a wrong geometry before you try to explain its pretty wake.

An unsettled result is still useful for illustrating a diagnostic, but its displayed mean must remain provisional. Automatic extension can help sampling without certifying a physically correct steady state. If all cases drift, changing device geometry may be less urgent than establishing a reliable baseline. If only a fine case oscillates, investigate whether the mesh reveals a more unsteady flow state or a numerical instability; neither explanation is proven by the oscillation alone.

Distinguish direct force from induced body change

For a wing case, compute the change in total load and compare it with the integrated wing load. The remainder belongs to body, wheels, and other exposed surfaces, subject to the integration's consistency checks. For a diffuser replacement, the baseline and variant body surfaces differ, so interpreting a separate “diffuser force” is ambiguous unless corresponding surface regions are defined. Compare total body load and matched floor regions rather than subtracting arbitrary part IDs.

Track the sign of total lift. A modification may add downforce relative to the baseline while the final car still lifts. Track the reference moment origin. A pressure snapshot can show where a mechanism occurs but cannot be integrated as though it were exactly the time-averaged force history. If a run lacks a field, retain

the quantitative record and say the flow explanation is not yet checked.

Evidence limits are part of the course

The raw examples establish that the exercise geometry can be run and inspected with EasyCFD. They do not establish acceptable accuracy for wings, canards, diffuser stall, or vehicle balance. The workbook intentionally contains no “correct CFD coefficient” to memorise. Your correct answer is a defensible statement about the evidence, its uncertainty, and the next discriminating test. Keep the difference between functional browser validation and aerodynamic qualification explicit in any report.

▼ **Day 3 checkpoint: a variant improves downforce by 15 N and your window spread is 30 N. What do you report?**

The comparison is unresolved by that temporal evidence, and additional uncertainties may be larger. Report the signed means, their window behaviour, and a repeat/refinement plan instead of declaring a useful gain.

Pitching moment, axle load, and tyre grip

You will derive equivalent aerodynamic axle loads and distinguish those loads from a complete handling prediction.

The location of the load matters

A car with 800 N downforce concentrated at the rear behaves differently from one with the same total split evenly. To calculate equivalent front/rear aero loads, use the total vertical force and the pitching moment about a declared reference. This is a static force-and-moment decomposition. The loads describe the aerodynamic increment that the axles must balance; they are not necessarily the pressure integrals on the front and rear halves of the body.

Let s be distance aft of the front axle and ℓ the wheelbase. Let N be positive downward load. Define M_{load} about the front axle as the moment that produces positive rear equivalent load. For vertical point loads $M_{\text{load}} = \sum s_i N_i$; include any additional aerodynamic couple with the same sign convention. Then $N_{\text{rear}} = M_{\text{load}}/\ell$ and $N_{\text{front}} = N_{\text{total}} - N_{\text{rear}}$. In EasyCFD's +X-aft, +Z-up frame, the appropriate Y moment about a road-level front-axle origin also includes hD from drag at height h . Always transform a saved moment to the chosen origin before applying the axle calculation.

$$N_{\text{rear}} = M_{\text{load, front axle}}/\ell$$

$$N_{\text{front}} = N_{\text{total}} - N_{\text{rear}}$$

$$M_{O2} = M_{O1} - (r_{O2} - r_{O1}) \times F$$

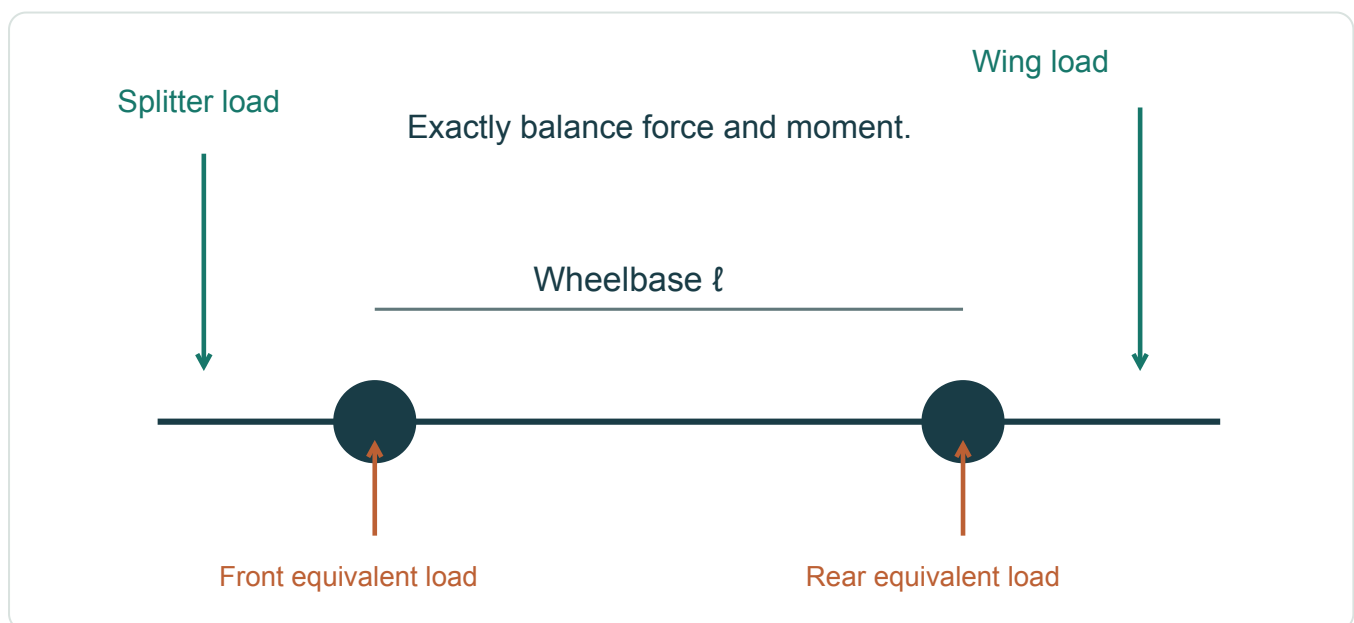


Figure 24. Equivalent axle loads for vertical forces and a moment. A wing behind the rear axle can unload the front while adding total downforce.

Workbench · Two vertical point loads and axle balance

Teaching statics: splitter and wing vertical point loads only, no drag-height moment, aerodynamic couple, suspension deflection, braking, or lateral load transfer. Negative front aero load means aerodynamic unloading.

Wheelbase	2.6 m	Splitter downward load	150 N
Splitter position aft of front axle	-0.4 m	Wing downward load	500 N
Wing position aft of front axle	3 m		

650.0 N

Total positive downforce

96.2 N

Front equivalent aero load

553.8 N

Rear equivalent aero load

1,440.0 N·m

Rear-loading moment

14.8%

Front fraction

0.000 N

Force closure

Equivalent aero loads are statics increments, not a tyre-grip or handling prediction.

Worked example

Wheelbase is 2.6 m. A splitter adds 150 N at $s = -0.4$ m, and a wing adds 500 N at $s = 3.0$ m. Total $N = 650$ N, $M_{\text{load}} = -60 + 1500 = 1440$ N·m, $N_{\text{rear}} = 553.8$ N, and $N_{\text{front}} = 96.2$ N. Without the splitter, the wing alone creates 576.9 N rear increment and -76.9 N front increment. More total downforce does not imply more front grip.

Front fraction is useful only with a meaningful total

Aero balance $\beta = N_{\text{front}}/N_{\text{total}}$ is commonly quoted as a percentage. If total downforce is near zero, the ratio becomes unstable. If one axle experiences lift, the fraction can be below 0% or above 100%; that is a warning about the physical load distribution, not a reason to clip the value. Report the signed axle increments as well as any percentage. Choose a desired balance from vehicle behaviour and tyre capability, not automatically from static mass distribution.

Add static weight, then include dynamics separately

For a 1100 kg car with 52% static front weight, total weight is 10,787 N; front static load is about 5610 N and rear about 5178 N. Add the equivalent aero increments at the evaluated speed to estimate quasi-static axle normal loads. Braking transfers approximately $m \times h_{\text{CG}}/l$ between axles in a simple rigid-body model. Cornering also produces left/right transfer influenced by track width, suspension, roll stiffness, and geometry. Do not merge these mechanisms into a single “aero kilograms” number.

Tyres are load sensitive: lateral force capability generally grows less than proportionally with normal load over a practical range, and combined braking/cornering uses a shared traction capability. Extra downforce can help but does not give a proportional increase in cornering speed. Temperature, camber, pressure, slip

angle, load transfer, and tyre construction all matter. Without an actual tyre model, course calculations establish load changes and hypotheses, not lap-time or handling guarantees.

Lab 10 · 30 min · Build a balance argument

Use the workbench to choose splitter and wing loads for a declared front fraction, then move the wing behind the rear axle. Explain the effect on front load. In EasyCFD, inspect detected axle positions and the saved moment origin before using the balance panel; verify wheel geometry and axle assignments. Compare the baseline and one package using signed equivalent front/rear loads where available.

Deliverable: an equilibrium check showing front + rear = total and rear × wheelbase = transformed moment. If moment reconciliation or axle metadata is missing, report the limitation instead of inventing a balance percentage.

DAY 4 / 20 / VEHICLE INTEGRATION

Speed, power, braking, and circuit tradeoffs

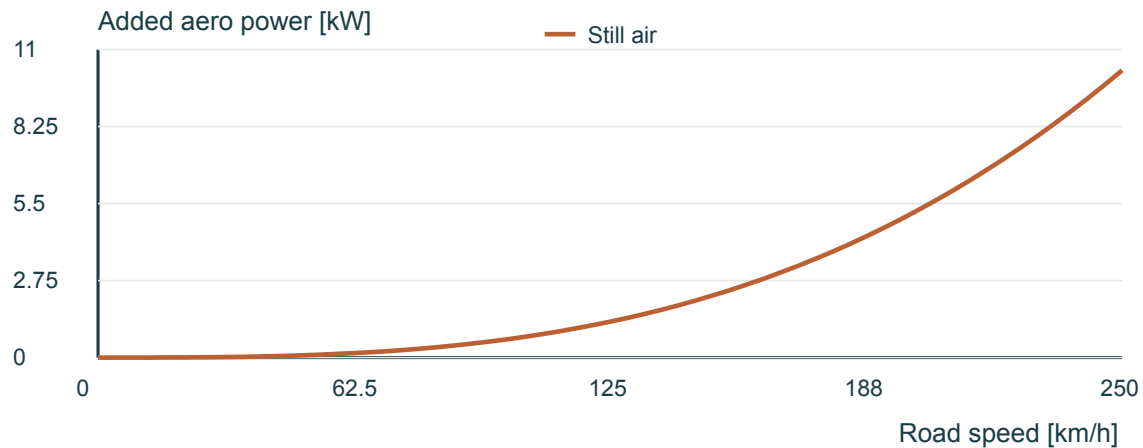
You will quantify the power cost of drag and explain why a useful aero package depends on the circuit and operating speed.

The straight has a budget

In still air, aerodynamic power at the wheels is $D \times \text{vehicle speed}$. For constant C_dA it grows with U^3 . At 200 km/h, adding 100 N drag costs about 5.56 kW of wheel power. Powertrain efficiency determines the engine or electrical-input cost. Rolling resistance, gradients, drivetrain losses, and acceleration add further demands. Use the correct available wheel power and gearing rather than treating engine headline power as a direct aerodynamic budget.

A steady top-speed estimate equates available wheel power to aero power plus other losses at that speed. Ignoring other losses and assuming fixed power and coefficients gives $U_{\max} \propto (P/C_dA)^{1/3}$. A 10% increase in drag area therefore reduces this simplified top speed by about 3.1%. This is a sensitivity calculation, not a prediction of a specific car's terminal speed, because gearing and power curves can change the outcome.

Added power from $\Delta C_d A = 0.05 \text{ m}^2$



Includes aerodynamic power only; assumes unchanged coefficient and density.

Figure 25. Added aerodynamic power for a constant added drag area of 0.05 m^2 in still air, $\rho = 1.225 \text{ kg/m}^3$. This is an analytic curve.

Downforce helps where speed and tyre demand coincide

Aero load grows with airspeed squared if the coefficients remain constant. In a slow hairpin, mechanical grip may dominate; in a fast corner, downforce can materially increase normal load. A circuit with long flat-out straights can punish drag, while a circuit with many fast corners can reward downforce. That tradeoff depends on the car's power, tyres, mass, balance, and driver limits. The same package is not optimal everywhere.

For a highly simplified constant- μ tyre and an evenly balanced car, lateral acceleration capacity might scale like $\mu(mg + N)/m$. Real tyres are load sensitive and axle-limited, so this formula often overstates the benefit. Even the simple model has speed coupling: N depends on the corner speed being solved for. An added wing that unloads the limiting front axle may give less benefit than its total downforce suggests.

Braking and pitch complicate the simple picture

Aerodynamic drag assists deceleration but is not a controllable tyre brake and falls with speed. Downforce can increase tyre braking capacity, yet longitudinal weight transfer and pitching motion change axle loads and floor clearances. A floor that is excellent in a level static pose can lose rear load in heavy braking or scrape at the front. A robust brake-zone package therefore needs aerodynamic pose data and a vehicle dynamics model or measured behaviour, not just a maximum-downforce setting.

Wind separates vehicle speed from airspeed. Drag force is set by the flow relative to the car; mechanical power expended by the moving car is the force component opposing vehicle motion times road speed. Do not multiply drag magnitude by total relative airspeed in an arbitrary crosswind and call it wheel power. At yaw, project the aerodynamic vector along the travel direction. The simple force calculator explicitly uses still air to avoid this ambiguity.

Use a transparent decision table

For each package list $\Delta C_d A$, front/rear downforce-area changes, operating range, mass, clearance cost, and uncertainty. Convert to forces at representative slow, medium, and fast speeds, then to still-air drag power. Check whether the useful load occurs at the speeds where the tyres need help. If a small drag saving is not resolved numerically, do not let a precise lap-time model turn it into a convincing fictional advantage.

Worked example

Take an illustrative package with $\Delta C_d A = 0.05 \text{ m}^2$ and $\Delta C_n A = 0.40 \text{ m}^2$. At 100 km/h it adds 23.6 N drag and 189 N downforce, costing 0.66 kW. At 200 km/h it adds 94.5 N drag and 756 N downforce, costing 5.25 kW. The incremental ratio is 8, but axle distribution and ride-height robustness still determine whether the package is useful.

▼ **Check: the package has the best downforce/drag ratio. Must it have the best lap time?**

No. Absolute load, axle balance, speed distribution, power, tyre response, robustness, mass, and constraints also matter. A ratio is one descriptor, not a lap-time result.

DAY 4 / 21 / DESIGN METHOD

Interactions, experiment design, and Pareto choices

You will quantify non-additivity and choose feasible candidates without relying on a single weighted score.

Devices interact through the shared flow field

A splitter can change floor entry. Skirts can alter underfloor leakage. A diffuser changes the rear wake. A rear wing can alter exit pressure and body separation. The complete package's change is therefore not generally the sum of isolated-device changes. Define an interaction effect for any metric Y using $IAB = Y_{AB} - Y_A - Y_B + Y_0$, where 0 is baseline, A and B are separate modifications, and AB is their combination. Use the same physical and numerical conditions for all four cases.

For positive downforce, an illustrative baseline of 100 N, A of 180 N, B of 220 N, and AB of 340 N gives interaction 40 N. That is more load than the additive expectation of 300 N. But the 40 N matters only if the four-case comparison is resolved within uncertainty. Apply the same formula to drag, axle loads, and moments. A beneficial downforce interaction can accompany a harmful drag or balance interaction.

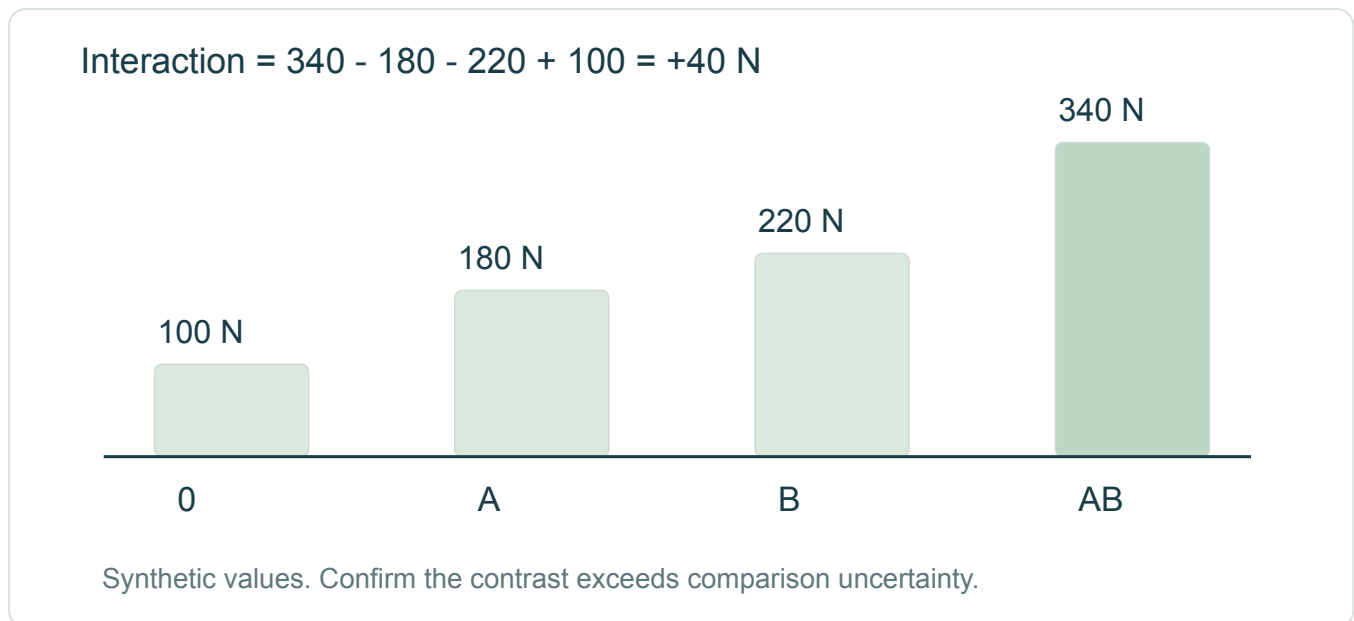


Figure 26. The four-case interaction contrast. Values are a synthetic worked example; the combined case must be run independently.

Screen factors, then study promising interactions

A one-factor-at-a-time sequence is easy to interpret but can miss interactions. A full two-level factorial study with k factors needs 2^k cases: four factors produce sixteen configurations before repeats. Add a centre point when meaningful to detect curvature and repeat key cases to estimate numerical variation. A fractional design reduces runs but aliases some effects; document which interactions you cannot separate. Do not infer a nonlinear optimum from two levels alone.

Choose factors that correspond to a physical mechanism: wing incidence, floor entry clearance, skirt gap, diffuser geometry, or outlet position. Keep unrelated geometry unchanged. Define each level precisely in millimetres and degrees. Randomising physical test order helps avoid temperature, wind, and tyre-state drift; in deterministic CFD the analogous protection is consistent settings plus repeats across grids and numerical conditions. A response surface should be checked with independent cases before being used to optimise.

Multiple objectives lead to a frontier

A candidate is Pareto dominated if another feasible candidate has no worse drag and no less useful downforce, with at least one strict improvement. The nondominated set is a frontier of tradeoffs; it is not a single winner. Balance, cooling, clearance, load-path, and uncertainty constraints can eliminate an apparent aerodynamic winner. A package with more rear load but insufficient front load may be infeasible even when its total downforce is high.

Workbench · Feasible tradeoffs

Synthetic design candidates chosen for teaching. These are not the paper's coefficients, not the course CFD outputs, and not recommendations. Assume a baseline with 200 N positive downforce. ΔD and ΔN are changes from that baseline; Front is each complete candidate's total front aero fraction, including the baseline load. Change the drag-power and front-balance limits to see which candidates remain feasible.

Added power ceiling at 200 km/h **6 kW** Minimum front aero fraction **40%**

Synthetic candidates evaluated at 200 km/h

Candidate	ΔD [N]	ΔN [N]	Front [%]	Status
Baseline	0	0	50	Feasible frontier
Small front package	25	180	64	Feasible frontier
Balanced package	70	520	47	Feasible frontier
Rear-heavy wing	100	760	25	Outside constraints
Large balanced package	150	850	44	Outside constraints
Inefficient package	135	500	47	Outside constraints

Changing a constraint changes the candidate set. A frontier still needs operating-envelope and uncertainty checks.

A weighted score such as $N - \lambda D$ can select a point for a declared value of λ , but weights hide priorities and their units. Show the frontier first, then state the operational reason for choosing a candidate. If uncertainty regions overlap strongly, retain several candidates or gather more evidence. Do not turn numerical noise into a highly precise optimum.

Lab 11 · 45 min · Measure an interaction

Use B0, splitter, canards, and splitter + canards from Lab 05, or create a four-case diffuser/wing study with body replacements handled correctly. Compute IAB for C_d , positive downforce coefficient, and front/rear load if available. Plot each package's force tradeoff and flag convergence warnings. Use the course's complete-package case only as a broader example; it cannot identify which pair caused an interaction.

Deliverable: a four-case interaction table and a decision that includes uncertainty and balance. If the contrast is unresolved, propose the smallest additional experiment that could settle it.

DAY 4 / 22 / VEHICLE INTEGRATION

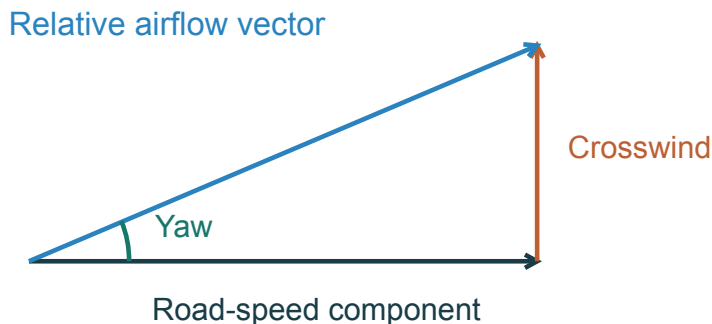
Yaw, pitch, and a robust aero map

You will design an operating envelope and distinguish road speed, relative wind, and geometric pose.

Crosswind is a vector problem

Let road-relative vehicle velocity be V_{car} and ambient wind velocity be W in a common ground frame. Relative air velocity in the car frame is $W - V_{car}$; its opposite convention can be used for oncoming wind if stated consistently. For a car travelling at 100 km/h with a perpendicular 5 m/s wind, the oncoming airspeed is $\sqrt{(27.78^2 + 5^2)} \approx 28.22$ m/s and yaw magnitude is $\text{atan}(5/27.78) \approx 10.2^\circ$. The road still moves directly rearward at 27.78 m/s in the car frame.

Side force, yawing moment, and rolling moment matter in this case. Endplates, exposed wheel faces, skirts, and asymmetric separation can create sensitivity that a zero-yaw study misses. Positive and negative yaw are not redundant for an asymmetric car, asymmetric cooling openings, or a numerically symmetry-constrained setup. Inspect both. A mean yawing moment alone does not define gust response without the vehicle dynamics and time-dependent aerodynamic behaviour.



Road stays longitudinal; inlet can have a lateral component.

Figure 27. Road motion and relative airflow differ under crosswind. The wind triangle is a teaching calculation.

Workbench · Relative wind triangle

Vector calculation for longitudinal headwind and lateral crosswind. The app's single speed/yaw input must be audited before claiming independent road and airspeed conditions.

Road speed	100 km/h	Headwind (negative = tailwind)	0 m/s
Lateral crosswind	5 m/s		
27.78 m/s Road motion / rolling speed		28.22 m/s Relative airspeed	
10.20° Yaw magnitude and sign		487.9 Pa Relative-air dynamic pressure	

The moving road has no lateral wind component. Wheel rotation follows the 27.78 m/s road-relative motion.

In the current WebGPU setup, the speed field sets the longitudinal inlet component U_x and the road speed; yaw adds $U_y = U_x \tan(\text{yaw})$. For example, 100 km/h with 10.2° yaw gives approximately 5 m/s lateral relative wind, while the road remains at 27.78 m/s. The resulting total airspeed is about 28.22 m/s. Check the saved freestream and dynamic pressure when comparing coefficients. This coupling can express pure crosswind, but an independent longitudinal headwind needs separate road and inlet speeds; that is part of the operating-map feature request.

Pose changes the complete installation

Pitch during braking moves the front splitter toward the road and changes the underfloor entry/rake. Rear squat changes diffuser clearance and wing/body incidence. Roll changes left/right floor and skirt gaps. Steering changes wheel exposure and wheelhouse flow. Suspension motion, tyre deflection, and chassis flexibility determine the actual pose. A rigid car translated vertically with rotating wheels fixed in the body is a different experiment from a sprung body moving relative to wheel centres.

Create pose variants in CAD with clear reference points and wheel-road geometry held appropriately. Record front and rear loaded ride height, pitch, roll, steering, and tyre treatment. A dedicated app tool for these states would help, but until it exists manual variants must be carefully inspected. Do not use the course's clearance slider as proof of a physically correct suspension-pitch study.

A modest first robustness matrix

Begin with the nominal case, $\pm 5^\circ$ and $\pm 10^\circ$ yaw, several feasible front/rear heights, and at least one braking-like and acceleration-like pose. Add combinations near suspected attachment transitions. If this becomes too costly, screen with the simpler solver and then qualify the promising and worst cases with an appropriate higher-fidelity method or test. Keep the actual road/air velocity convention recorded, especially under crosswind.

Report minimum useful front load, maximum drag cost, balance excursions, and sensitivity gradients across the envelope. A peak-load setting that falls off abruptly with a 10 mm height change may be worse than a slightly lower-load broad plateau. Make the objective robust performance under realistic variation. Its

definition needs tolerances for geometry, alignment, ride height, wind, surface roughness, and measurement uncertainty.

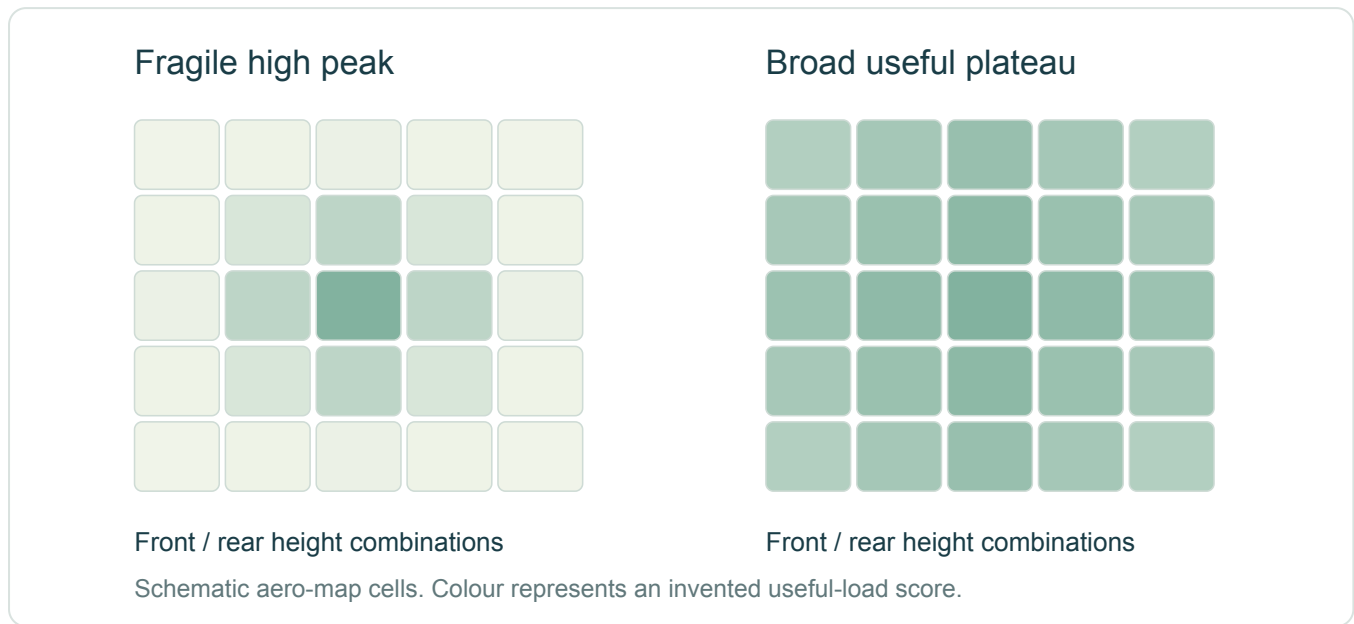


Figure 28. A schematic aero-map heatmap contrasts a narrow peak with a broad plateau. The cell colours are invented; they explain the study structure.

Feedback can couple aero and suspension

More downforce compresses springs, changing ride height, which changes downforce again. A simple static coupling solves suspension displacement and aero load together; dynamic coupling also needs damping, mass, road input, and time-dependent forces. Strong height sensitivity can create problematic behaviour even when a rigid nominal case looks excellent. A static RANS sweep cannot establish porpoising frequency or stability. Identify the potential feedback and obtain the needed dynamics evidence.

▼ Check: why might a nominally “best” low-clearance diffuser be a poor track choice?

It may lose load, shift balance, contact the road, or change flow state under realistic pitch, roll, yaw, and roughness. The worst relevant condition can matter more than the nominal peak.

DAY 4 / 23 / QUALIFICATION

Physical validation and a buildable modification

You will plan useful measurements and separate aerodynamic evidence from structural and operational readiness.

A test must discriminate between hypotheses

Choose a measurement that can confirm or refute the proposed mechanism. Pressure taps can show whether floor suction and diffuser recovery changed. Tufts and oil-flow methods can reveal surface-flow direction and possible separation. A force balance can measure aerodynamic loads under controlled wind-tunnel conditions. Suspension loads or deflections can indicate on-car load changes when mechanical effects are modeled and calibrated. No single method measures everything.

A rolling-road wind tunnel helps reproduce road-relative motion, but tyre shape, contact patch, support interference, blockage, scale, and Reynolds number still require attention. A fixed-floor small model may be useful for qualitative screening while giving misleading underfloor loads. State those limits. When comparing to CFD, reproduce geometry, pose, velocity, turbulence, road/wheel treatment, and the test facility where relevant; otherwise the mismatch combines numerical and test-condition differences.

On-car tests need control and repetition

Coastdown can help estimate road-load changes, but wind, slope, tyre temperature/pressure, drivetrain losses, and speed measurement confound the aerodynamic component. Use an established calibrated protocol on an appropriate controlled facility. Downforce cannot be inferred uniquely from a quicker lap or a driver's confidence. Separate driver variability, tyre evolution, fuel mass, traffic, weather, and mechanical setup. A-B-A sequences and repeated conditions help reveal drift but do not remove all confounding.

Instrumented ride height and suspension deflection require spring, motion-ratio, bump-stop, and damping interpretation. Pressure taps need calibrated sensors, suitable tubing response, and known locations. If the vehicle is pitching or rolling, static deflection-to-load conversion is insufficient. A measurement's bandwidth and uncertainty must suit the proposed effect. A 10 N change is not meaningful if the total measurement chain cannot resolve it.

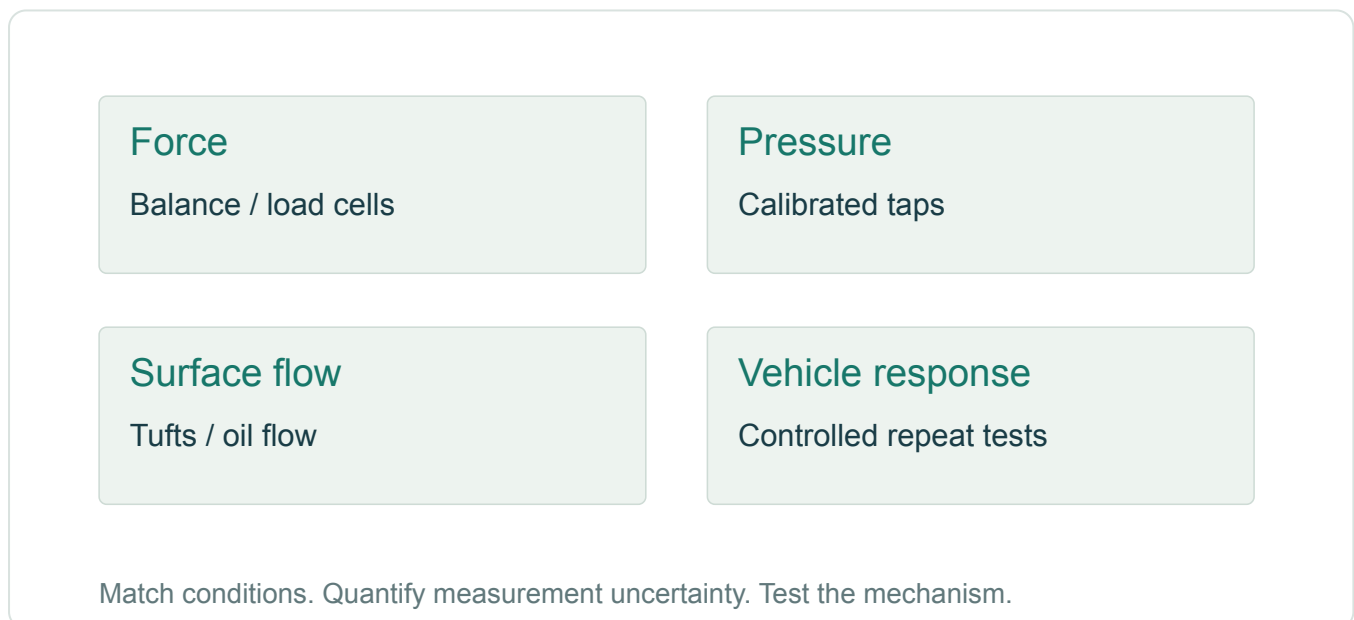


Figure 29. Independent evidence routes: force, pressure, surface flow, and vehicle response. A comparison must reproduce conditions and quantify measurement uncertainty.

Structure and packaging are separate checks

Define an aerodynamic load envelope with speed, wind, pose, and uncertainty, then transfer it to structural analysis. Include realistic distribution, load paths, fastener loads, fatigue, impact/contact, material degradation, and allowable deflection. A CFD force integral does not prove a bracket, composite laminate, boot lid, or bumper can carry that load. A geometry with free-floating wings and omitted mounts cannot qualify a real build.

A splitter needs ground-contact and debris considerations. A floor needs heat, drainage, service access, and ventilation. A diffuser must accommodate exhaust, suspension, and structural members. A wing needs mounting stiffness and an acceptable deflection-to-incidence relationship. Endplates and canards may be subject to dimensional or competition constraints. Check the actual applicable requirements for the intended use before release; this course does not prescribe jurisdiction-specific road legality or a ruleset.

Make a release decision with an evidence ladder

Separate concept plausibility, geometry readiness, numerical verification, physical validation, structural readiness, and operational readiness. Passing one rung does not imply the next. A prototype can be approved for a limited instrumented test without being approved for unrestricted use. Record which conclusion each artifact supports and who reviews the remaining work. This is normal engineering traceability, not an obstacle to experimentation.

Worked example

A simulated wing adds 200 N at 100 km/h. Constant-coefficient scaling suggests 800 N at 200 km/h, but the structural input should also account for the relevant off-design aerodynamic envelope and uncertainty. "The simulation ran" is evidence of software execution; "the mount passed" requires an independent structural analysis and suitable test.

▼ Check: you gained 0.8 s on one lap after fitting canards. Did you measure their aerodynamic benefit?

No. That observation combines many variables and does not isolate aero force or mechanism. Use controlled repeated comparison and targeted measurements to establish the change.

DAY 4 / 24 / CAPSTONE

Capstone: a defensible aerodynamic package

You will turn a design brief into a reviewable decision with explicit evidence and remaining work.

Your assignment

Design a candidate package for the car and use you chose in Lab 01. You may use the teaching car or an adequately documented car model. Select a primary goal, a drag or power budget, an axle-balance requirement, and an operating envelope. Every numerical target is a project requirement you choose and justify, not a general recommendation from this course. Use at least one front device and one rear/floor device, and retain the baseline as a candidate.

Do not attempt to optimise every geometric parameter in four hours. Pick a small hypothesis set: for example, a splitter creates useful front load, a wing recovers rear load, and skirts improve a coherent floor. Use a staged experiment: baseline; individual changes; one combination; then a robustness and verification subset. A well-explained unresolved result is a better engineering deliverable than an unsupported optimum.

Lab 12 · 4 hours · Prepare the design review

1. **Define, 30 min:** brief, constraints, coordinate/force convention, reporting area, and three testable hypotheses.
2. **Prepare, 45 min:** geometry manifest, groups, exact one-body/one-wing checks, dimensions, road/wheel settings, numerical plan, and output naming.
3. **Evaluate, 75 min:** run the most discriminating cases or audit the included evidence; collect histories, paired differences, and matched flow views. Record any failed or unsettled case.
4. **Challenge, 45 min:** assess refinement/domain sensitivity, interaction, balance, and one off-design condition. State if evidence is insufficient.
5. **Review, 45 min:** produce a short recommendation, an evidence table, and the next physical and structural tests.

The four-hour capstone uses the earlier labs as its data foundation. Longer solver runs and real-car geometry preparation may require additional project time.

The review package

Begin with the concrete decision: retain baseline, develop candidate A, or gather more evidence. Show the acceptance criteria and whether each is met, unresolved, or failed. Include total dimensional drag and downforce at a stated speed, signed front/rear aero increments, reference axes/area/origin, and solver qualification. Use absolute differences as well as percentages, especially near zero lift or drag changes.

Include one table of geometry variants and run settings, one history plot, two matched flow views that test your mechanism, and a force/moment comparison. Link the raw exports and geometry recipe so another engineer can reconstruct the comparison. Separate the new device's load from changes on the existing body where that decomposition is valid. Do not cherry-pick a late iteration or hide an unresolved run behind the strongest picture.

Assess the package as a system

Explain whether separate changes were additive and what caused any interaction. Discuss speed scaling, power demand, axle balance, clearance and yaw sensitivity, cooling implications, and a credible structural load path. State which of those claims are calculations, simulations, observations, or assumptions. If the peak package is fragile across the operating range, a more moderate candidate may be worth developing.

Your final paragraph should name the next discriminating experiment and its acceptance gate. For example: “Before selecting between the wing variants, repeat the installed cases with qualified near-wall resolution and matched physical pressure/load measurements; accept a ranking only if the difference exceeds uncertainty and front load remains above the project requirement across the specified yaw/pose envelope.” That is a concrete engineering decision even when today's evidence is inconclusive.

Self-review rubric · 100 points

Area	Points	What good work contains
Physics	20	Correct signs, pressure/normal reasoning, mechanism, and scaling limits.
Experiment	20	Controlled settings, geometry provenance, independent comparisons.
Verification	20	History, grid/domain sensitivity, uncertainty, and failures retained.
Vehicle integration	20	Balance, speed/power, operating envelope, cooling and load paths.
Decision	20	Clear conclusion, reproducible evidence, and a targeted next test.

▼ Final checkpoint: what does “I understand car aerodynamics” mean operationally?

You can formulate a mechanism, predict its direction under stated assumptions, test it with controlled conditions, interpret the force and flow evidence, quantify what remains uncertain, and decide which next test or design action follows. Memorising a preferred wing or diffuser angle is not enough.

APPENDIX A / WORKED CHECKS

Exercises with worked answers

Attempt these without the answers first. The numbers are teaching inputs, not fitted results for any particular vehicle. Show units, force signs, area convention, and assumptions in every calculation.

1. One speed, two reference areas

A run at 120 km/h has $D = 550 \text{ N}$ in air at 1.225 kg/m^3 . Calculate C_d using $A_{\text{ref}} = 2.0 \text{ m}^2$ and 2.3 m^2 . What changed physically?

▼ Worked answer

$U = 33.333 \text{ m/s}$, $q = 680.56 \text{ Pa}$. C_d is $550/(680.56 \times 2.0) = 0.4041$ or $550/(680.56 \times 2.3) = 0.3514$. The force did not change. The difference is only the reporting denominator. $C_d A = 0.8082 \text{ m}^2$ in both cases.

2. Added downforce while still lifting

Baseline $C_l = +0.25$; variant $C_l = +0.10$ at $A_{\text{ref}} = 2.2 \text{ m}^2$, 100 km/h. Calculate the added downforce and final lift.

▼ Worked answer

$qA = 472.61 \times 2.2 = 1039.74 \text{ N}$. Added downforce = $-(0.10 - 0.25) \times 1039.74 = 155.96 \text{ N}$. Final lift remains $+103.97 \text{ N}$. The variant improves vertical load but does not create net downforce.

3. Wing reporting convention

A wing has $S = 0.50 \text{ m}^2$, $C_{l,\text{wing}} = -1.2$, $C_{d,\text{wing}} = 0.15$. Convert its contribution to car-area coefficients with $A_{\text{car}} = 2.2 \text{ m}^2$, assuming the same q .

▼ Worked answer

$C_{l,\text{car}} \text{ contribution} = -1.2 \times 0.50/2.2 = -0.2727$, $C_{d,\text{car}} \text{ contribution} = 0.15 \times 0.50/2.2 = 0.03409$. These are the wing's force contribution only, not the whole-car changes. Body interaction can change the final delta.

4. Scale-model Reynolds number

A 4 m car at 30 m/s has $\nu = 1.5 \times 10^{-5} \text{ m}^2/\text{s}$. What Re does a 1 m model have at the same speed? What speed would match full-size Re in the same air?

▼ Worked answer

Full size Re = 8 million; model Re = 2 million. Matching requires 120 m/s. At a $\approx 343 \text{ m/s}$ that is Mach 0.35, so simply increasing speed creates a different compressibility regime and practical facility demands. Use a documented similarity strategy rather than claiming exact equivalence.

5. Ideal diffuser expansion

A constant-width ramp has $h_1 = 0.10 \text{ m}$, $\ell = 0.80 \text{ m}$, $\theta = 7^\circ$. Given prescribed $U_1 = 35 \text{ m/s}$ and $\rho = 1.225$, compute ideal exit speed and pressure recovery.

▼ Worked answer

$h_2 = 0.10 + 0.80 \tan 7^\circ \approx 0.1982 \text{ m}$; $R \approx 1.982$. $U_2 \approx 17.66 \text{ m/s}$. $q_1 \approx 750.31 \text{ Pa}$; ideal recovery $\approx 750.31(1 - 1/1.982^2) \approx 559 \text{ Pa}$. These values assume uniform attached flow and the prescribed mass flow. They do not predict floor pressure or car downforce.

6. Three-grid arithmetic

Independent comparable grids give fine $C_d = 0.3000$, medium = 0.3060, coarse = 0.3195 with spacing ratio $r = 1.5$. Calculate p , extrapolated C_d , and fine GCI.

▼ Worked answer

Difference ratio = $0.0135/0.006 = 2.25$, so $p = \ln(2.25)/\ln(1.5) = 2$. $\phi_{\text{ext}} = 0.3000 - 0.006/1.25 = 0.2952$. Fine GCI = $1.25(0.006/0.3000)/1.25 = 0.020 = 2.0\%$. This depends on monotonic asymptotic refinement and converged cases; it is not total physical uncertainty.

7. Aero couple without vertical force

A pure aerodynamic moment adds +260 N·m in the rear-loading convention about the front axle of a 2.6 m wheelbase car. Total vertical aerodynamic force is zero. What are the equivalent axle increments?

▼ Worked answer

$N_{\text{rear}} = 260/2.6 = +100$ N, $N_{\text{front}} = -100$ N. There is no net vertical aero load, but front is unloaded and rear is loaded. The front fraction of total downforce is undefined because the denominator is zero.

8. Drag at height

A rearward drag force of 200 N acts at height 0.8 m with no net vertical force. Using a road-level front-axle origin and the EasyCFD axes, what equivalent axle-load change follows for $\ell = 2.6$ m?

▼ Worked answer

The Y moment is $hD = +160$ N·m. $N_{\text{rear}} = +61.54$ N and $N_{\text{front}} = -61.54$ N in the declared convention. This is part of aerodynamic force/moment decomposition. Actual vehicle equilibrium under propulsion/braking also includes those forces, their application heights, and inertial effects; do not interpret the aero increment as the complete dynamic load transfer.

9. A four-case interaction

Positive downforce coefficients are 0.10 baseline, 0.18 with A, 0.22 with B, and 0.31 with AB. Compute the interaction and its force at 100 km/h, $A = 2.2$ m².

▼ Worked answer

$IAB = 0.31 - 0.18 - 0.22 + 0.10 = +0.01$. Dimensional interaction is $0.01 \times 1039.74 \approx 10.40$ N. If variation in the paired cases is larger than that, the interaction is unresolved.

10. A yawed inlet

A car travels at 120 km/h in a 6 m/s perpendicular ambient wind. Find relative airspeed and yaw magnitude. What road velocity should a car-fixed moving-ground boundary represent?

▼ Worked answer

Airspeed is $\sqrt{(33.333^2 + 6^2)} \approx 33.87$ m/s, yaw $\approx \text{atan}(6/33.333) = 10.20^\circ$. Road speed is 33.333 m/s directly rearward, with no lateral component. Wheel rolling speed follows road motion, not the airspeed magnitude.

11. A drag-power budget

Added drag area is 0.040 m². Find added still-air power at 200 km/h in density 1.225 kg/m³. Compare it with a project limit of 3 kW.

▼ Worked answer

$q \approx 1890.43$ Pa, $\Delta D \approx 75.62$ N, $\Delta P = 75.62 \times 55.556 \approx 4.20$ kW. The candidate exceeds that stated project limit. Lower-speed performance or downforce benefit does not make the constraint disappear; revise the package or explicitly revise the requirement.

12. An honest uncertainty statement

Cd means are 0.350 and 0.346. Window bands are ± 0.006 and ± 0.005 , and grid sensitivity has not been tested. Write your conclusion.

▼ Worked answer

"The observed drag difference is -0.004, but it is not resolved by the available temporal evidence and numerical sensitivity is unquantified. Longer sampling and paired refinement/domain checks are needed before claiming a gain." Do not combine the displayed bands as independent statistical standard deviations; they are window-spread diagnostics.

APPENDIX B / FIELD GUIDE

Glossary and formula sheet

Aerodynamic balance	Distribution of equivalent aerodynamic vertical load between axles, based on force and pitching moment.
Angle of attack / incidence	Angle between a section reference chord and its incoming air; a mounting angle need not equal local incidence.
Aspect ratio	b^2/S ; span/chord for a rectangular wing. Declared using wing planform geometry.
Attachment	Boundary-layer state in which flow continues along a surface rather than forming an extended separated region.
Boundary layer	Near-wall region with strong velocity gradients influenced by viscosity and turbulent transport.
Blockage	Projected body area divided by tunnel cross-section; a confinement indicator, not an accuracy certificate.
Camber	Curvature of an airfoil's mean line.
Canard / dive plane	Small front-corner aero surface with direct load and possible downstream flow effects.
Cd / Cl	Drag and signed lift coefficients based on q and a declared reference area.
Centre of pressure	Effective force location for a chosen force/moment decomposition; unstable when the relevant force approaches zero.
Cp	$(p - p_\infty)/q_\infty$, a local static-pressure coefficient.
Diffuser	Expanding passage intended to recover static pressure; its flow depends on the complete inlet-to-exit system.
Downforce	Positive downward aerodynamic load, defined in this course as $-L$.
Drag area	$C_d A$, in m^2 ; useful across different area conventions.
Dynamic pressure	$q = \frac{1}{2}\rho U^2$, in pascals.
Endplate	Surface near a wing tip that changes spanwise flow and loading; it also adds drag and yaw sensitivity.
Flow pass	In the browser solver, a car-length/freestream-speed scale used for simulation effort; local stepping is pseudo-time.
GCI	Grid convergence index, a discretisation-uncertainty reporting method under stated refinement assumptions.
Ground effect	Aerodynamic changes due to proximity and interaction with the ground.
Gurney flap	Small trailing-edge tab used to modify load and pressure recovery.
Induced drag	Drag associated with a finite lifting system's induced flow; not all drag on a wing.
Laminar / turbulent	Different flow regimes of transport and fluctuations; neither word alone establishes attachment.

LES / RANS / URANS	Different turbulence modeling/resolution strategies; their predictions require their own qualification.
Mach number	U/a , the speed ratio relative to sound.
Moment origin	Reference position used for torque integration. Changing it changes the reported moment by a force-arm term.
Pareto frontier	Nondominated feasible candidates in a multiple-objective comparison.
Pressure recovery	Increase in static pressure as flow decelerates; a demanding recovery can provoke separation.
Rake	Longitudinal floor inclination associated with front/rear ride-height difference.
Reynolds number	UL/ν ; compare using a declared characteristic length.
Separation	Detachment of a boundary layer; diagnose with surface-flow and off-wall evidence.
Splitter	Front horizontal surface that can produce downforce from a pressure difference.
Streamline / pathline / streakline	Instantaneous field tangent / one particle trajectory / particles released from a source. They coincide only in steady flow.
Total pressure	For the limited incompressible relation, static plus dynamic pressure; losses distinguish wakes from reversible acceleration.
Wall function	Model relating near-wall mean flow and stress without resolving every inner-layer scale.
y+	$u\tau/\nu$, a solution-dependent nondimensional wall distance.
Yaw	Relative-flow angle in plan view; ambient wind and road motion must be separated.

$$\begin{aligned}
 U \text{ [m/s]} &= \text{speed [km/h]} / 3.6 \quad ; \quad \text{kgf} = N / 9.80665 \\
 q &= \frac{1}{2}\rho U^2 \quad ; \quad Re = UL/\nu \quad ; \quad M = U/a \\
 D &= q C_d A \quad ; \quad N = -q C_l A \quad ; \quad P = D U \text{ (still air)} \\
 AR &= b^2/S \quad ; \quad Di \approx N^2/(\pi e q b^2) \text{ (isolated attached wing)} \\
 R &= [h + \ell \tan \theta]/h \quad ; \quad \text{ideal recovery}/q_{throat} = 1 - 1/R^2 \\
 N_{rear} &= M_{load}/\ell \quad ; \quad N_{front} = N_{total} - N_{rear} \\
 IAB &= YAB - Y_A - Y_B + Y_O
 \end{aligned}$$

APPENDIX C / REPRODUCIBILITY

Your lab record and reading notes

Use one record per comparison. Record design/version, geometry hashes or generator revision, enabled body and wing, speed, yaw, pose, density, reference area, road and wheel conditions, turbulence input, domain bounds, engine/version, cell count, resolution, wall/layer treatment, averaging window, grid identifier, force signs, moment origin, Cd/Cl, dimensional forces, axle loads, temporal bands, warnings, field-view scales, and interpretation. Record failures and missing outputs as such. “Same defaults” is less durable than a settings snapshot.

A useful four-column lab narrative

Prediction	Controlled conditions	Observation	Interpretation / next test
What mechanism and sign do you expect?	What varied, and what stayed fixed?	What do history, forces, and matched fields show?	What does the evidence support? What could disprove it?

The interactive HTML includes a local notebook and text export. Browser-specific notes are not embedded in this published PDF.

Critical EasyCFD improvements identified while writing this course

The course works with the existing app and reports missing capabilities explicitly. Existing accuracy qualification and complete force/report issues are linked below. New course-driven issues cover reproducible operating-map/sweep studies with independent road/wind and pose conditions, mutually exclusive replacement groups, cooling duct/porous-core modeling, and STL face-orientation diagnostics. These are future feature requests; follow the links for concrete behaviour and acceptance evidence.

- #19 · Reproducible aero maps with static poses and independent road/wind inputs
- #20 · Mutually exclusive replacement-body and wing variant groups
- #21 · Cooling ducts and radiator pressure loss with mass-flow evidence
- #22 · Diagnose inconsistent STL face winding before running CFD
- #16 · Complete aerodynamic accuracy qualification for lift, wings and advanced runs (existing issue)
- #15 · Complete force/moment tables and offline simulation reports (existing issue)

APPENDIX D / SOURCES AND ATTRIBUTION

Sources, assumptions, and further study

This is an original course with original diagrams, teaching examples, calculator equations, and generated synthetic geometry. It does not reproduce third-party textbook chapters or unlicensed images. External links are further reading and focused references. All numerical plots identify whether their values are analytic, synthetic, published, or actual course-run output. Sources were checked on 7 October 2026.

1. Guerrero, Castilla, and Eid (2022). A Numerical Aerodynamic Analysis on the Effect of Rear Underbody Diffusers on Road Cars. Applied Sciences 12(8), 3763. DOI 10.3390/app12083763. CC BY 4.0. Table 3 transcribed with attribution; ratios/deltas independently recomputed. Study conditions are taken from Sections 2-4. Author-university PDF mirror.
2. Senior and Zhang (2001). The force and pressure of a diffuser-equipped bluff body in ground effect. Journal of Fluids Engineering 123, 105-111. DOI 10.1115/1.1340637. The repository abstract provides the limited experimental observation of load loss at very low ride height; exact thresholds are not generalised.
3. Zerihan (2001). An investigation into the aerodynamics of wings in ground effect. University of Southampton research thesis. Further experimental study of wing/ground interactions, not a polar for the course wing.
4. NASA Glenn. Dynamic pressure. Definition reference for q ; course force/power calculations are original.
5. NASA Glenn. Bernoulli's equation. Reference for restrictions of the simple relation; no claim that it predicts viscous car flow.
6. NASA Glenn. Boundary layer. Introductory context for near-wall flow and separation.
7. NASA Glenn. Induced drag coefficient. Isolated finite-wing model; the assumed workbench polar is not fitted to car data.
8. NASA Glenn. What is drag? Terminology context; no universal car drag budget is imported.
9. NASA/NPARC. Examining spatial (grid) convergence. Further reading for observed order and GCI assumptions; the course's worked sequence is synthetic.
10. NASA Turbulence Modeling Resource. Primary reference hub for model definition and verification cases.
11. OpenFOAM. Force coefficients. Coefficient/origin/direction configuration; match documentation to installed version.
12. OpenFOAM. yPlus. Near-wall diagnostic documentation.
13. EasyCFD validation report and repository. Current app capabilities, known numerical limitations, and solver provenance; review the version associated with your run.

A next reading path

After the course, study vehicle-aerodynamics references that cover experimentally validated ground effect, wheel interactions, cooling drag, and aero/vehicle-dynamics coupling. Work through the Southampton experimental material and the NASA verification resources above, then reproduce one documented benchmark with its original geometry and boundary conditions. A benchmark reproduction teaches more than comparing unrelated C_d headlines.

EasyCFD · Car aerodynamics course · Edition 1 · Original course/geometry under the repository MIT license. Published-paper data retain their CC BY attribution. PDF and HTML share the same source; interactive controls are printed at their stated default settings.